



Acoustics of porous media

Lecture 3: The Biot theory

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- Frame is not motionless !
- General overview on the model
- Conservative poroelastic material
 - Lagrangian approach
 - Propagation equation
 - Boundary conditions
- Introduction of dissipation
- Different representations
- Biot waves
- Simple problems
- Extended Biot models
 - Nonlinear Biot theory
 - Double porosity materials



M.A Biot
(1905-1985)

Reprinted from THE JOURNAL OF THE ACOUSTICAL SOCIETY OF AMERICA, Vol. 28, No. 1, 348-378, March, 1956
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Theory of Propagation of Elastic Waves in a Fluid-Saturated Porous Solid. I. Low-Frequency Range

M. A. Biot^{*}
Shell Development Company, RCA Building, New York, New York
(Received September 1, 1955)

A theory is developed for the propagation of stress waves in a porous elastic solid containing a compressible viscous fluid. The approach of the present treatment is in materials where fluid and solid are of comparable densities as for instance in the case of water saturated rock. The paper devoted here as Part I is restricted to the lower frequency range where the assumption of Poiseuille flow is valid. The extension to the higher frequency will be treated in Part II. It is found that the material may be described by four nondimensional parameters and a characteristic frequency. There are two dilatational waves and one rotational wave. The physical interpretation of the results is clarified by treating first the case where the fluid is incompressible. The case of a material containing a viscous fluid is then developed and discussed separately. Phase velocity dispersion curves and attenuation coefficients for the three types of waves are plotted as a function of the frequency for various combinations of the characteristic parameters.

1. INTRODUCTION

OUR purpose is to establish a theory of propagation of elastic waves in a system composed of a porous elastic solid saturated by a viscous fluid. It is assumed that the fluid is compressible and may flow relative to the solid causing friction or shear. Part I which is presented here assumes that the relative motion of the fluid in the pores is of the Poiseuille type. As already pointed out by Kirchhoff this is valid only below a certain frequency which we denote by f_0 and which depends on the kinematic viscosity of the fluid and the size of the pores. Extension of the theory in the frequency range above f_0 will be presented in Part II. We have in mind particularly the application to cases where the fluid is a liquid, and we have therefore disregarded the thermodynamic effect. We include only materials such that the walls of the main pores are impenetrable and for which the pore size is concentrated around its average value. Extension to more general materials will be considered along with the thermodynamic effect in a later and more complete theory.

Development of the theory proceeds as follows. Section 2 introduces the concepts of stress and strain in the aggregate including the fluid pressure and dilatation. Relations are established between these quantities for static deformation in analogy with a procedure followed in the theory of elasticity for porous materials developed in reference 1. Section 3 considers the dynamics of the material when the fluid is assumed to be without viscosity. This case is of practical interest since this represents the limiting behavior of wave propagation at very high frequency. It introduces the concepts of apparent mass and dynamic coupling between fluid and solid. The wave propagation in the absence of friction is analyzed in Secs. 4 and 5. It is found that there is one rotational wave and two dilatational waves. A remarkable property is the possible existence of a wave such that no relative motion occurs between fluid and solid.

This is obtained if a "dynamic compatibility" relation is verified between the elastic and dynamic constants. This is a case where dissipation due to fluid friction will disappear. In Sec. 6 the dynamic relations are derived with the addition of fluid viscosity and Sec. 7 derives the properties of the propagation of the waves when dissipation is present. It is found that the phase velocity of rotational waves increases slightly with the frequency f while the absorption coefficient is proportional to the square of the frequency. All plots are presented nondimensionally by referring to a characteristic frequency f_0 which depends on the kinematic viscosity of the fluid and the pore diameter. The characteristic frequency f_0 may be considered as a frequency scale of the material; the nondimensional distances of the plots being the ratio f/f_0 . There are two dilatational waves denoted as waves of the first and second kind. The waves of the second kind are highly attenuated. They are in the nature of a diffusion process, and the propagation is closely analogous to heat conduction. The waves of the first kind are true waves. The dispersion is practically negligible with a phase velocity increasing or decreasing with frequency depending on the mechanical parameters. The absorption coefficient is proportional to the square of the frequency as for the rotational waves. In cases close to the dynamic compatibility condition, the dispersion and attenuation of the waves of the first kind tend to vanish. The attenuation of this wave may therefore vary widely for materials of similar composition and may be larger or smaller than the attenuation of the rotational waves.

A beginning along the present line was made by Frenkel.² He discusses the rotational and dilatational waves, but the subject is not treated in detail and important features are neglected.

Sound absorption in material containing air was the subject of extensive work by Zwicker and Kosten.³ Rotational waves are not considered, and simplified equations are used for the dilatational waves. The

^{*} Consultant.

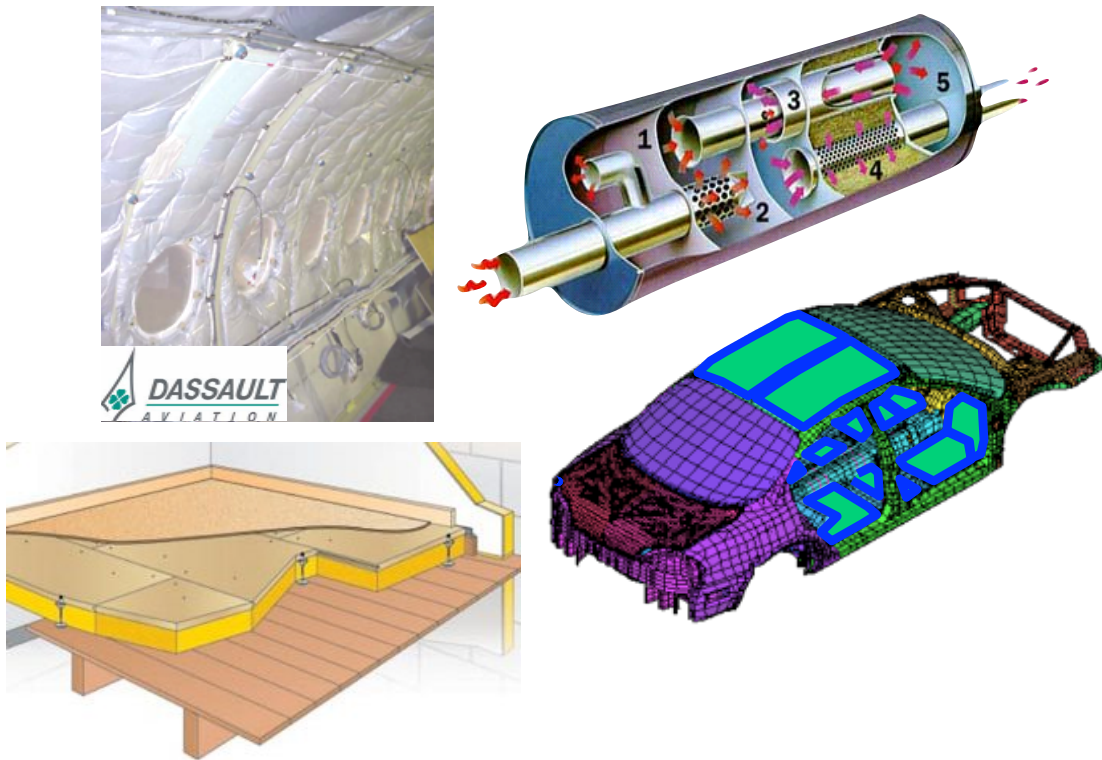
^{*} M. A. Biot, J. Appl. Phys. 26, 102 (1955).

¹ J. Frenkel, J. Phys. 16, 1, 5, 61 & 120 (1949).

² J. Zwicker and C. W. Kosten, Sound Absorbing Materials (Ottawa Publishing Company, Inc., New York, 1949).

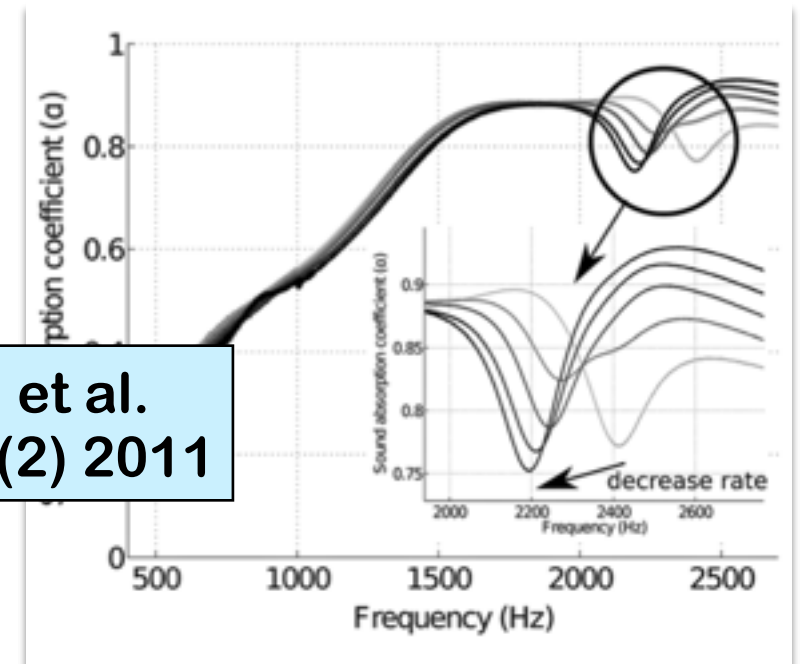
Is frame motion important ?

Mechanical excitation



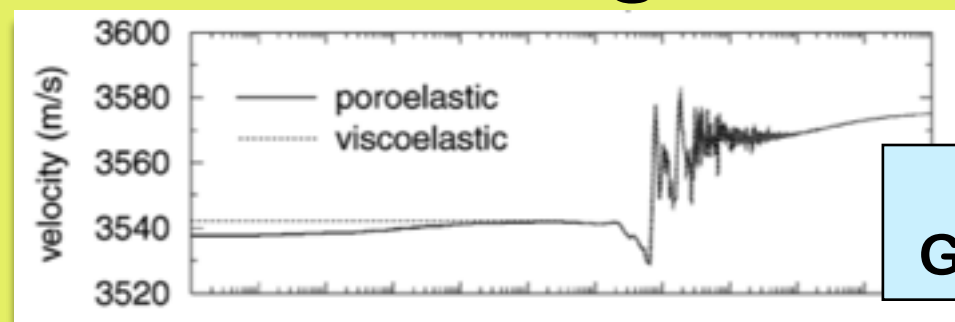
Airborne excitation

Geslain et al.
JASA 130(2) 2011



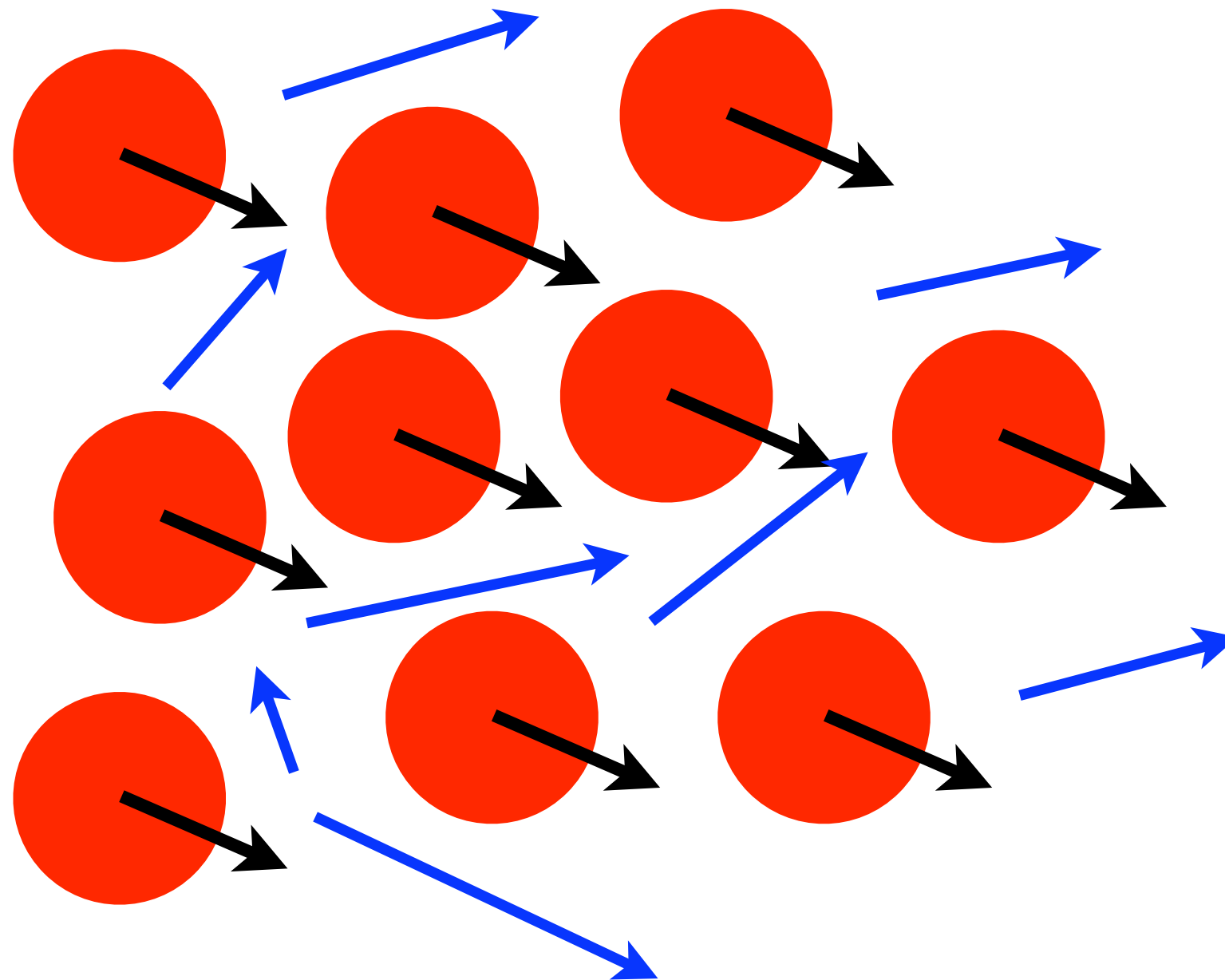
Quarter wavelength
of the solid phase

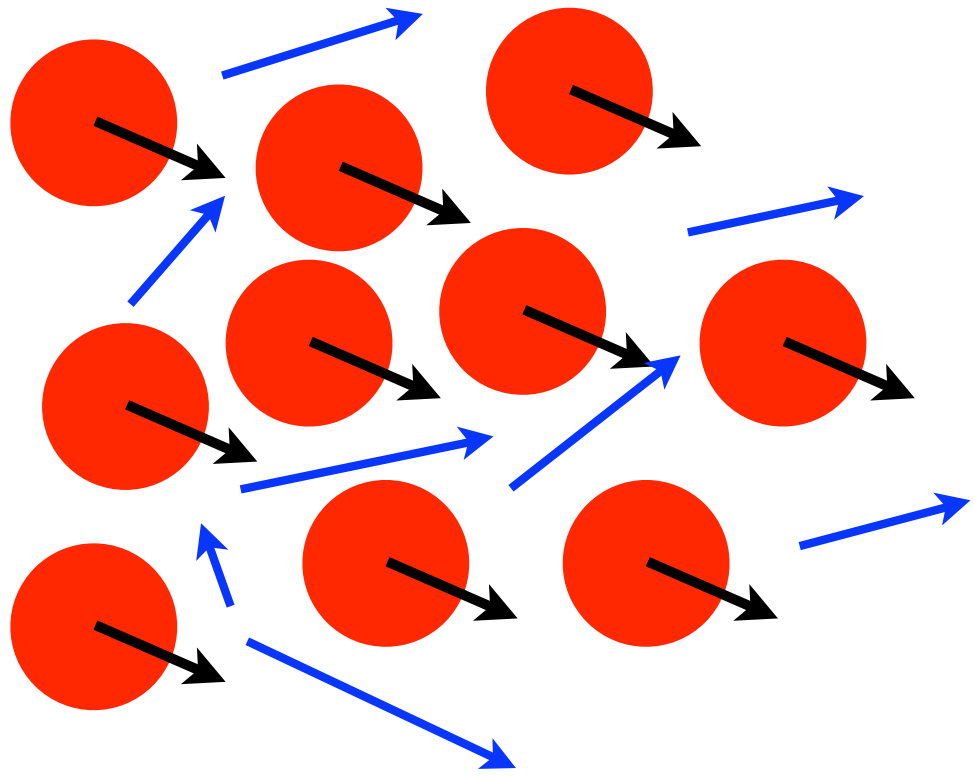
Origin of the Biot theory



Pride et al.
Geophysics 67(1) 2002

Geotechnical applications
(Heavy fluid)



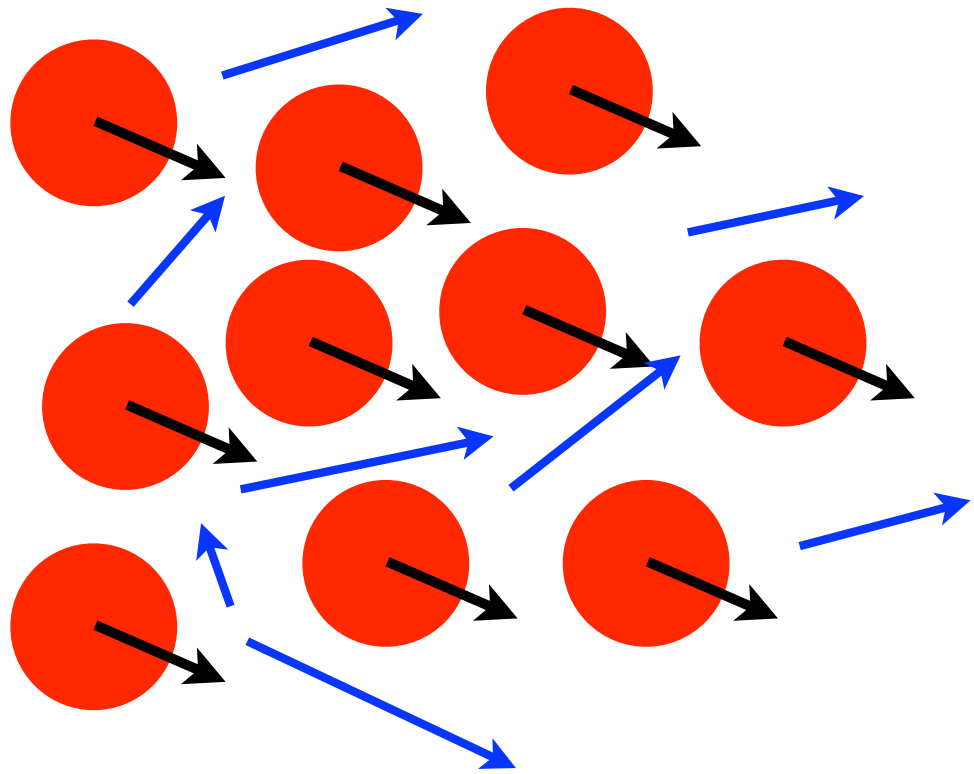


Consequences

- Conservation of the fluid topology at microstructure scale
- Fluid properties

$$\tilde{\rho}_{eq}, \quad \tilde{K}_{eq}$$

- At macro scale: No shear stress in the fluid



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For the moment

- Material is supposed as conservative (No dissipation)
- Solid phase is supposed elastic
- Poroelastic material (PEM)

Solid displacement

$$\mathbf{u}^s = \langle u_\mu^s \rangle_{\Omega_s}$$

Fluid displacement

$$\mathbf{u}^f = \langle u_\mu^f \rangle_{\Omega_f}$$

Objective:

- ✓ Motion equation
- ✓ Constitutive laws
- ✓ Propagation equations

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✓ Approach 1 (classical mechanics)

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Which density, stress ?

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Which density, stress ?

Chosen by Biot, presented in this lecture

$$\mathbf{q} = \{q_i(x_1, x_2, x_3, t)\}, \quad i = 1..n$$

Fields of representation

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Fields of representation

Stationarity of action

$$\delta \mathcal{S} = 0 \quad \mathcal{S} = \int_{t_1}^{t_2} \mathcal{L} \left(\mathbf{q}, \dot{\mathbf{q}}, \frac{\partial \mathbf{q}}{\partial x_i}, t \right) dt$$

Topics on least action principle

$$\mathbf{q} = \{q_i(x_1, x_2, x_3, t)\}, \quad i = 1..n$$

Fields of representation

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Euler-Lagrange's equation

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_j} + \sum_{i=1}^3 \frac{d}{dx_i} \frac{\partial \mathcal{L}}{\partial \frac{\partial q_j}{\partial x_i}} - \frac{\partial \mathcal{L}}{\partial q_j} = 0$$

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We want:

- ✓ Wave like equations: terms up to first derivative only
- ✓ Homogeneous: no dependence on time and space
- ✓ Isotropic: rotation symmetry
- ✓ Linear equations: Quadratic forms

$$\mathbf{q} = \mathbf{u} = \{u_1, u_2, u_3\} \quad \mathcal{L} \left(\dot{\mathbf{u}}, \frac{\partial \mathbf{u}}{\partial x_i} \right) = ?$$

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We are looking for terms which:

- ✓ **verify the rotation symmetry**
- ✓ **do not disappear in Euler's Lagrange equations**

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$$\mathcal{L} \left(\dot{\mathbf{u}}, \frac{\partial \mathbf{u}}{\partial x_i} \right) = \underbrace{\alpha \dot{\mathbf{u}}^2}_{\mathcal{T}(\dot{\mathbf{u}})} + \underbrace{\beta \nabla \cdot \mathbf{u}^2 + \gamma (\nabla \wedge \mathbf{u})^2}_{\mathcal{V}(\frac{\partial \mathbf{u}}{\partial x_i})}$$

Kinetic energy

Strain energy

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Decomposition not
always valid

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Euler-Lagrange's equation

$$2\alpha \ddot{\mathbf{u}} + 2\beta \nabla \nabla \cdot \mathbf{u} + 2\gamma \nabla \wedge \nabla \wedge \mathbf{u} = 0$$

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$$2\alpha \ddot{\mathbf{u}} + 2\beta \nabla \nabla \cdot \mathbf{u} + 2\gamma \nabla \wedge \nabla \wedge \mathbf{u} = 0$$

$$\alpha = \rho$$

with “usual notations” $\beta = -(\lambda + 2\mu)$

$$\gamma = \mu$$

$$\rho \ddot{\mathbf{u}} = (\lambda + \mu) \nabla \nabla \cdot \mathbf{u} + \mu \nabla \wedge \nabla \wedge \mathbf{u}$$

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$$\rho \ddot{\mathbf{u}} = (\lambda + \mu) \nabla \nabla \cdot \mathbf{u} + \mu \nabla \wedge \nabla \wedge \mathbf{u} = \nabla \cdot \boldsymbol{\sigma}$$

$$\sigma_{ij} = -\frac{\partial \mathcal{L}}{\partial u_{i,j}} = \lambda \nabla \cdot \mathbf{u} \delta_{ij} + 2\mu \varepsilon_{ij} \quad \varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$$

Overview of lagrangian approach:

- ✓ Generality and abstraction
- ✓ Physics, necessary to interpret constants
- ✓ Motion equations
- ✓ Constitutive laws
- ✓ Here, difference of kinetic and strain energy density

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Question to answer:

For a two displacement fields medium, what is the most general form for the lagrangian density ?

$$\mathcal{L} \left(\dot{\mathbf{u}}^s, \dot{\mathbf{u}}^f, \frac{\partial \mathbf{u}^s}{\partial x_i}, \frac{\partial \mathbf{u}^f}{\partial x_i} \right) = \mathcal{T}(\dot{\mathbf{u}}^s, \dot{\mathbf{u}}^f) - \mathcal{V} \left(\frac{\partial \mathbf{u}^s}{\partial x_i}, \frac{\partial \mathbf{u}^f}{\partial x_i} \right)$$

6 fields $\implies$ 6 Euler's Lagrange equations

$$\mathcal{T} = \alpha (\dot{\mathbf{u}}^s)^2 + \beta (\dot{\mathbf{u}}^f)^2 + \gamma (\dot{\mathbf{u}}^s - \dot{\mathbf{u}}^f)^2$$

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Additional term

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Additional term

Interpretation of constants

$$\begin{aligned}\mathcal{T}(\mathbf{u}^f = 0) &= (\alpha + \gamma)(\dot{\mathbf{u}}^s)^2 \\ &= \left\langle \frac{1}{2} \rho_s (\dot{\mathbf{u}}_\mu^s)^2 \right\rangle_\Omega = \frac{1}{2} \underbrace{\rho_s (1 - \phi)}_{\rho_1} \underbrace{\langle (\dot{\mathbf{u}}_\mu^s)^2 \rangle_{\Omega_s}}_{(\dot{\mathbf{u}}^s)^2}\end{aligned}$$

$$\alpha + \gamma = \frac{\rho_1}{2}$$

$$\mathcal{T} = \alpha (\dot{\mathbf{u}}^s)^2 + \beta (\dot{\mathbf{u}}^f)^2 + \gamma (\dot{\mathbf{u}}^s - \dot{\mathbf{u}}^f)^2$$

Additional term

Interpretation of constants

$$\begin{aligned} \mathcal{T}(\mathbf{u}^s = 0) &= (\beta + \gamma)(\dot{\mathbf{u}}^s)^2 \\ &= \left\langle \frac{1}{2} \rho_0 (\dot{\mathbf{u}}_\mu^f)^2 \right\rangle_\Omega = \frac{1}{2} \underbrace{\rho_0 \phi}_{\rho_2} \underbrace{\langle (\dot{\mathbf{u}}_\mu^s)^2 \rangle_{\Omega_f}}_{\alpha_\infty (\dot{\mathbf{u}}^s)^2} \end{aligned}$$

$$\alpha + \gamma = \frac{\rho_1}{2}$$

$$\mathcal{T} = \alpha (\dot{\mathbf{u}}^s)^2 + \beta (\dot{\mathbf{u}}^f)^2 + \gamma (\dot{\mathbf{u}}^s - \dot{\mathbf{u}}^f)^2$$

Additional term

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$$\begin{aligned}\mathcal{T}(\mathbf{u}^s = \mathbf{u}^f) &= (\alpha + \beta)(\dot{\mathbf{u}}^s)^2 \\ &= \frac{1}{2} \left\langle \rho_s (\dot{\mathbf{u}}_\mu^s)^2 + \rho_0 (\dot{\mathbf{u}}_\mu^f)^2 \right\rangle_\Omega = \frac{1}{2} (\rho_1 + \rho_2) (\dot{\mathbf{u}}^s)^2\end{aligned}$$

$$\alpha + \gamma = \frac{\rho_1}{2} \quad \beta + \gamma = \frac{\rho_2 \alpha_\infty}{2} \quad \alpha + \beta = \frac{1}{2} (\rho_1 + \rho_2)$$

$$\mathcal{T} = \alpha (\dot{\mathbf{u}}^s)^2 + \beta (\dot{\mathbf{u}}^f)^2 + \gamma (\dot{\mathbf{u}}^s - \dot{\mathbf{u}}^f)^2$$

$$\rho_{11}\rho_{22} - \rho_{12}^2 > 0$$

Additional term

Interpretation of constants

$$\mathcal{T}(\mathbf{u}^s = \mathbf{u}^f) = (\alpha + \beta)(\dot{\mathbf{u}}^s)^2$$

$$= \frac{1}{2} \left\langle \rho_s (\dot{\mathbf{u}}_\mu^s)^2 + \rho_0 (\dot{\mathbf{u}}_\mu^f)^2 \right\rangle_\Omega = \frac{1}{2} (\rho_1 + \rho_2) (\dot{\mathbf{u}}^s)^2$$

$$\alpha + \gamma = \frac{\rho_1}{2} \quad \beta + \gamma = \frac{\rho_2 \alpha_\infty}{2} \quad \alpha + \beta = \frac{1}{2} (\rho_1 + \rho_2)$$

$$2\mathcal{T} = \rho_{11} (\dot{\mathbf{u}}^s)^2 + \rho_{22} (\dot{\mathbf{u}}^f)^2 + 2\rho_{12} \dot{\mathbf{u}}^s \cdot \dot{\mathbf{u}}^f$$

$$\rho_{11} = \rho_1 - \rho_{12}, \quad \rho_{22} = \rho_2 - \rho_{12}$$

$$\rho_1 = (1 - \phi)\rho_s$$

$$\rho_{12} = -\phi\rho_0(\alpha_\infty - 1) < 0$$

$$\rho_2 = \phi\rho_0$$

$$\begin{aligned} 2\mathcal{V} = & (A + N)(\nabla \cdot \mathbf{u}^s)^2 + R(\nabla \cdot \mathbf{u}^f)^2 + Q(\nabla \cdot \mathbf{u}^s \nabla \cdot \mathbf{u}^f) \\ & + N(\nabla \wedge \mathbf{u}^s)^2 + \gamma(\nabla \wedge \mathbf{u}^f)^2 + \gamma'(\nabla \wedge \mathbf{u}^s \cdot \nabla \wedge \mathbf{u}^f) \end{aligned}$$

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The fluid does not restore shear forces $\gamma = \gamma' = 0$

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Gedanken experiments to interpret constants

$$A = \frac{(1 - \phi)[1 - \phi - \frac{K_b}{K_s}]K_s + \phi \frac{K_s K_b}{K_f}}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}} - \frac{2N}{3}$$

$$Q = \frac{[1 - \phi - \frac{K_b}{K_s}]\phi K_s}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}}$$

$$R = \frac{\phi^2 K_s}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}}$$

- N : Shear modulus of the frame
- K_s : Bulk modulus of the skeleton material
- K_b : Bulk modulus of the frame
- K_f : Bulk modulus of the air in the pores



**Biot and Willis
JAM 24 1957**

$$\begin{aligned}\rho_{11} \frac{\partial^2 \mathbf{u}^s}{\partial t^2} + \rho_{12} \frac{\partial^2 \mathbf{u}^f}{\partial t^2} &= (A + N) \nabla \nabla \cdot \mathbf{u}^s + N \nabla^2 \mathbf{u}^s + Q \nabla \nabla \cdot \mathbf{u}^f \\ \rho_{12} \frac{\partial^2 \mathbf{u}^s}{\partial t^2} + \rho_{22} \frac{\partial^2 \mathbf{u}^f}{\partial t^2} &= Q \nabla \nabla \cdot \mathbf{u}^s + R \nabla \nabla \cdot \mathbf{u}^f\end{aligned}$$

$$\begin{aligned}\rho_{11} \frac{\partial^2 \mathbf{u}^s}{\partial t^2} + \rho_{12} \frac{\partial^2 \mathbf{u}^f}{\partial t^2} &= \nabla \cdot \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f) \\ \rho_{12} \frac{\partial^2 \mathbf{u}^s}{\partial t^2} + \rho_{22} \frac{\partial^2 \mathbf{u}^f}{\partial t^2} &= \nabla \cdot \boldsymbol{\sigma}^f(\mathbf{u}^s, \mathbf{u}^f)\end{aligned}$$

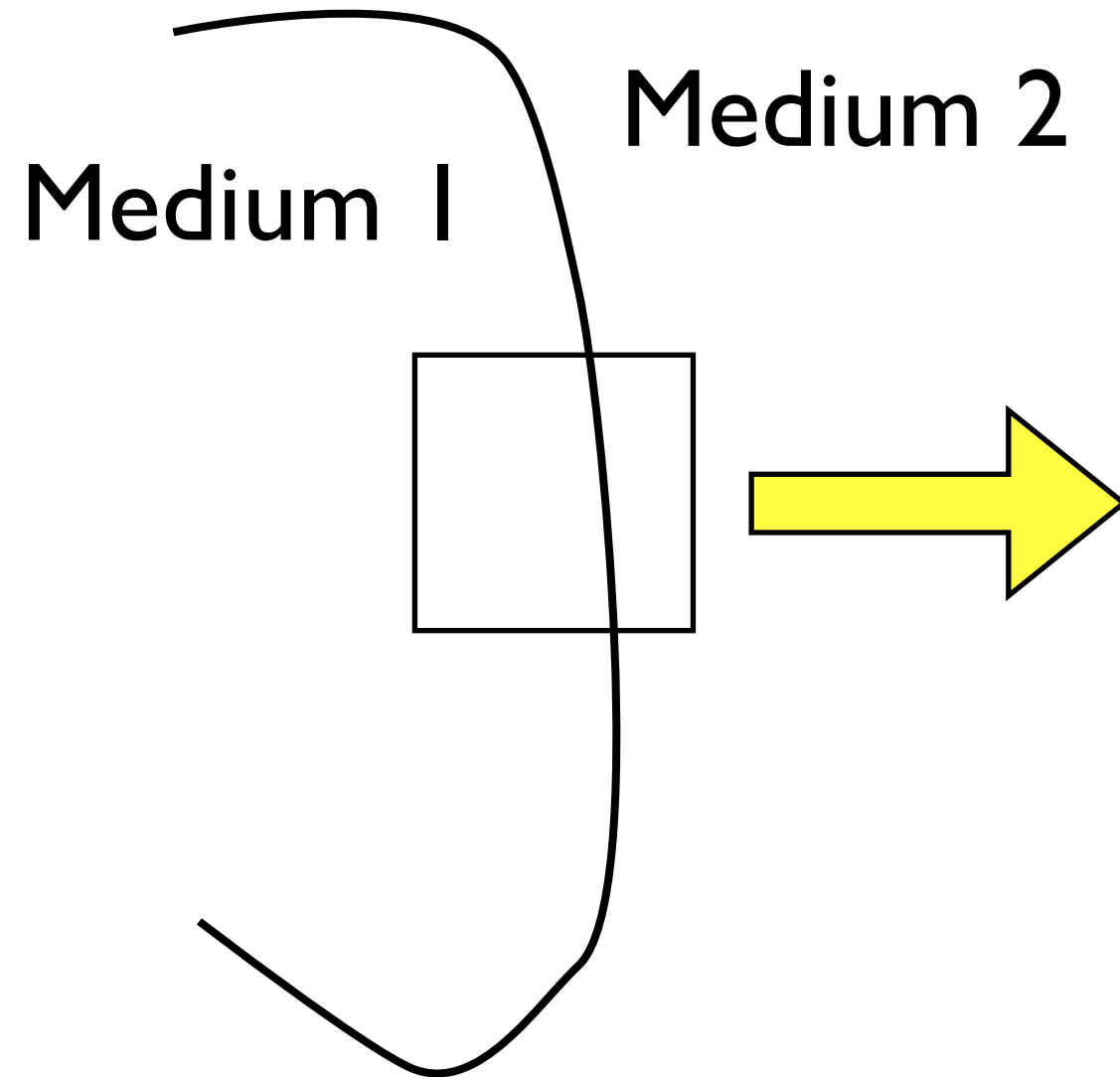
Partial stress tensor of the solid phase

$$\sigma_{ij}^s = -\frac{\partial \mathcal{L}}{\partial \mathbf{u}_{i,j}^s} = A \nabla \cdot \mathbf{u}^s \delta_{ij} + 2N \varepsilon_{ij}^s + Q \nabla \cdot \mathbf{u}^f \delta_{ij}$$

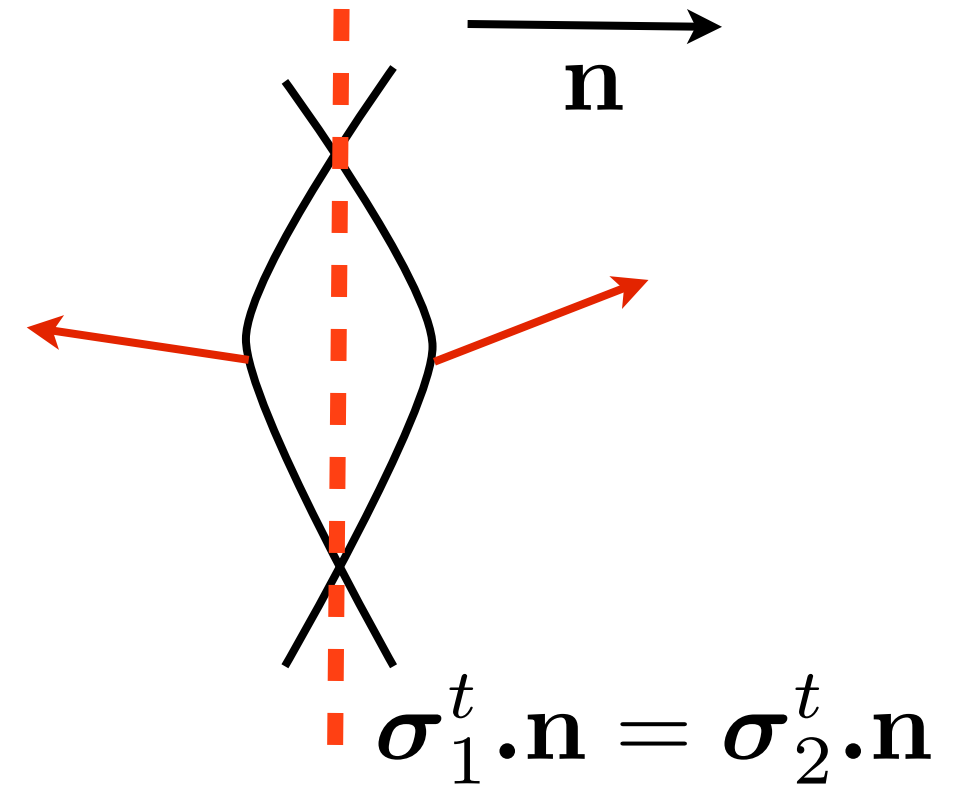
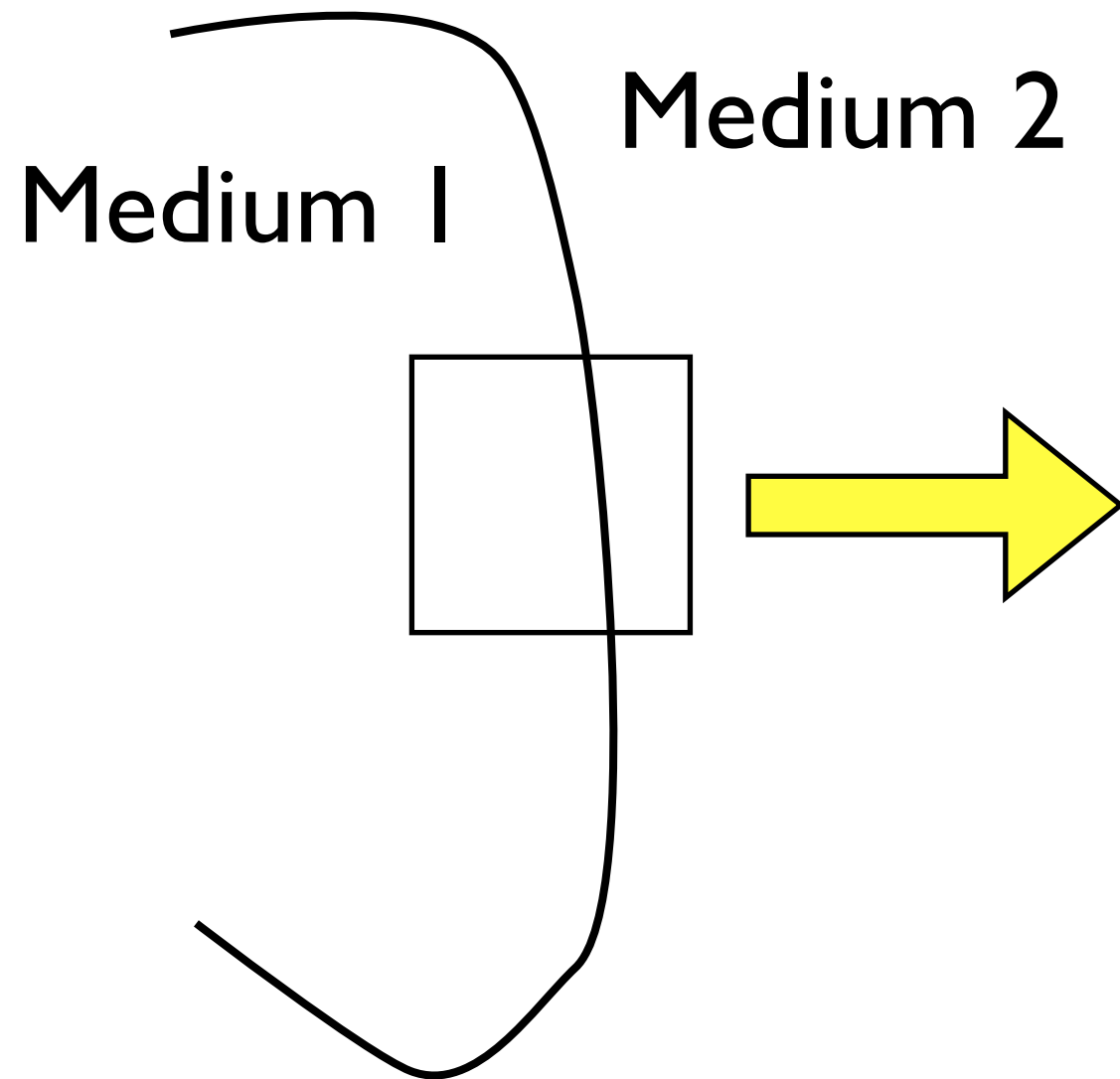
Partial stress tensor of the fluid phase

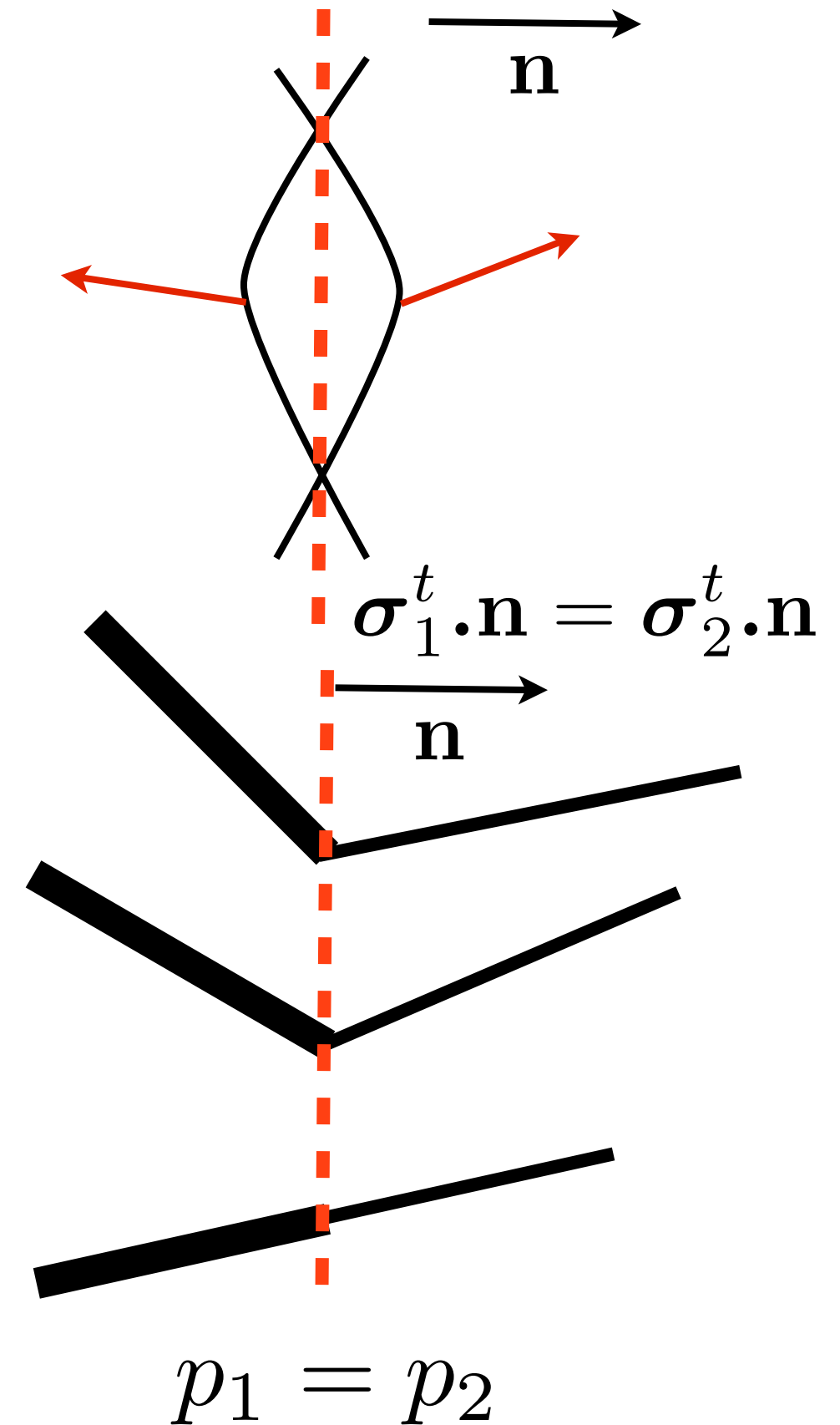
$$\sigma_{ij}^f = -\frac{\partial \mathcal{L}}{\partial \mathbf{u}_{i,j}^f} = Q \nabla \cdot \mathbf{u}^s \delta_{ij} + R \nabla \cdot \mathbf{u}^f \delta_{ij}$$

Stress continuity relations

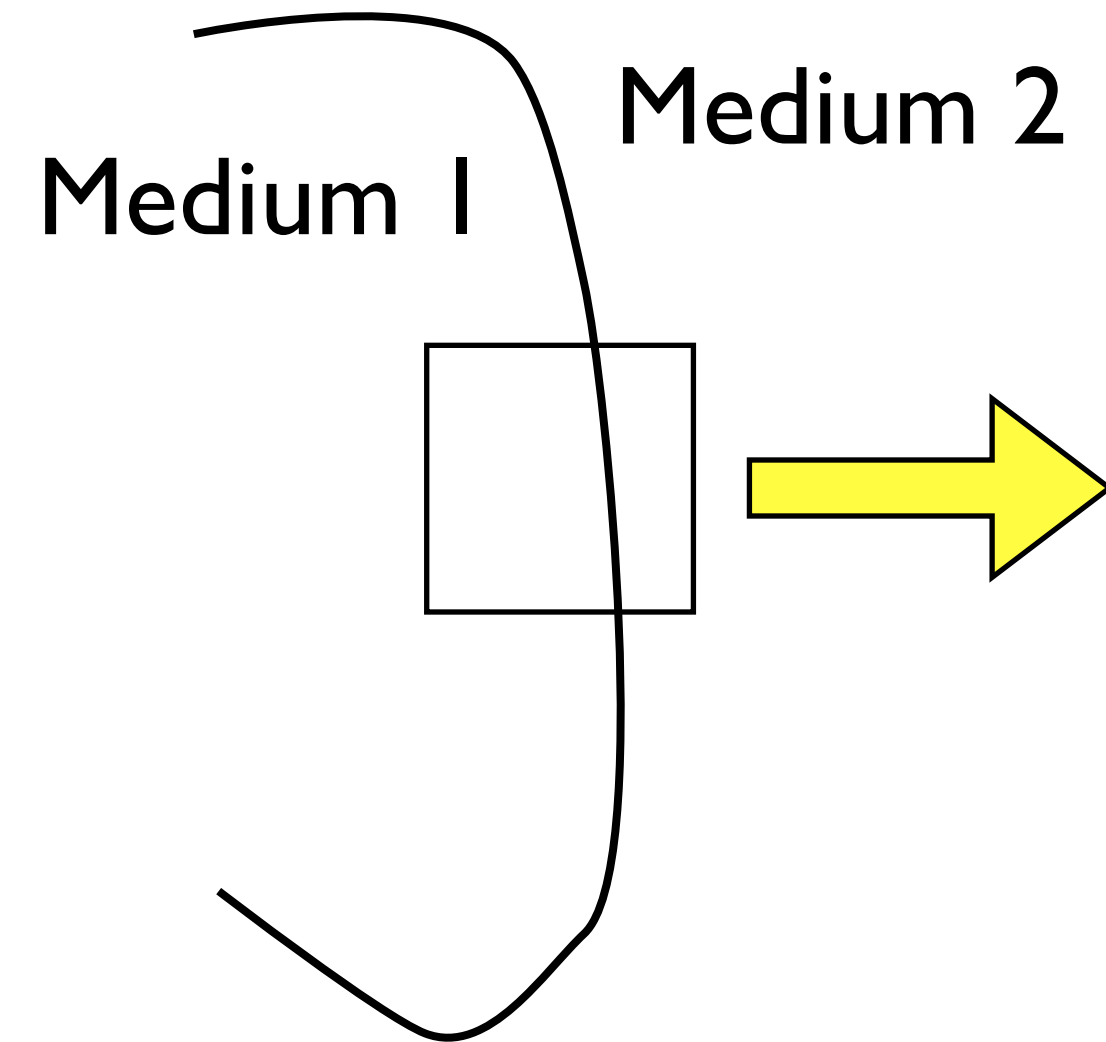


Stress continuity relations

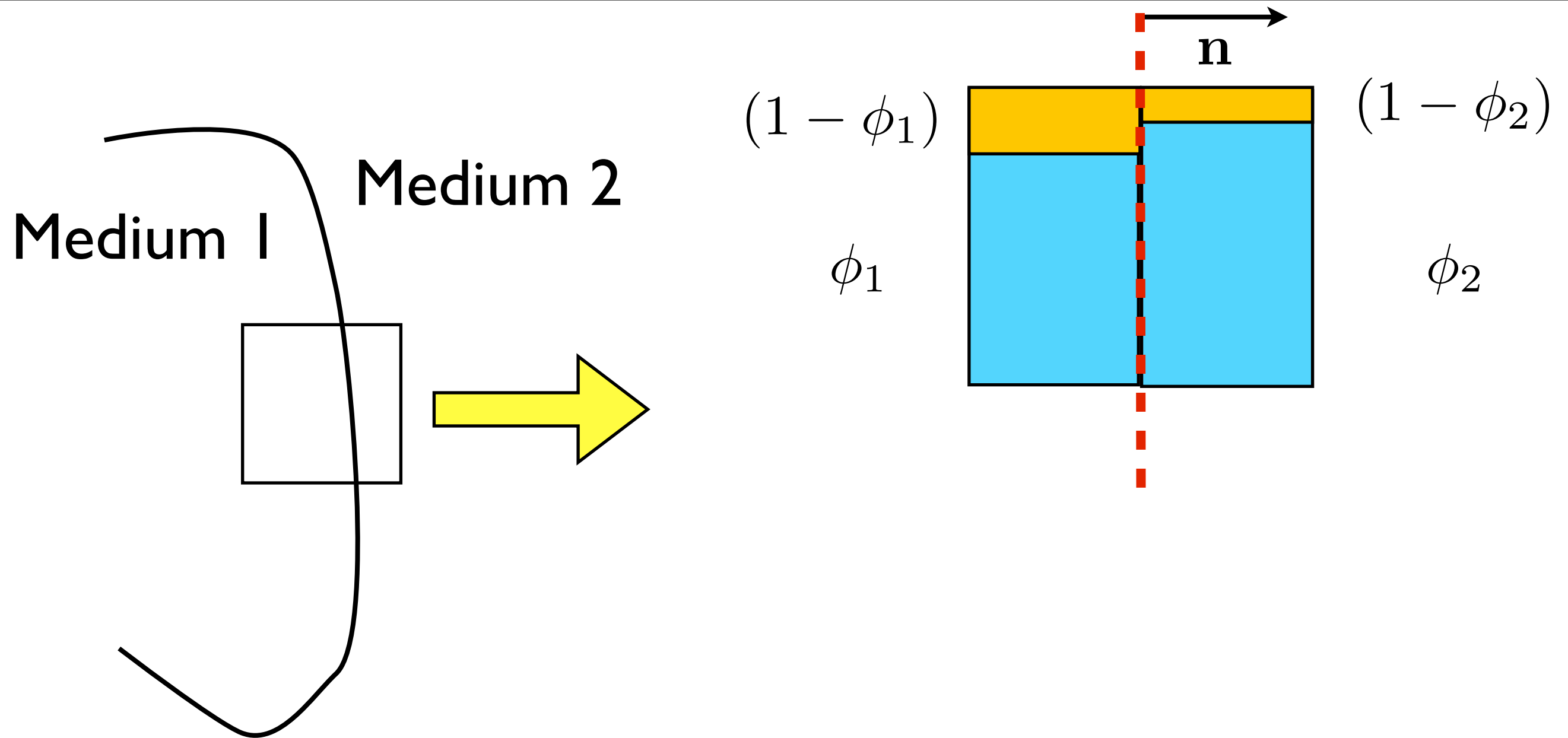




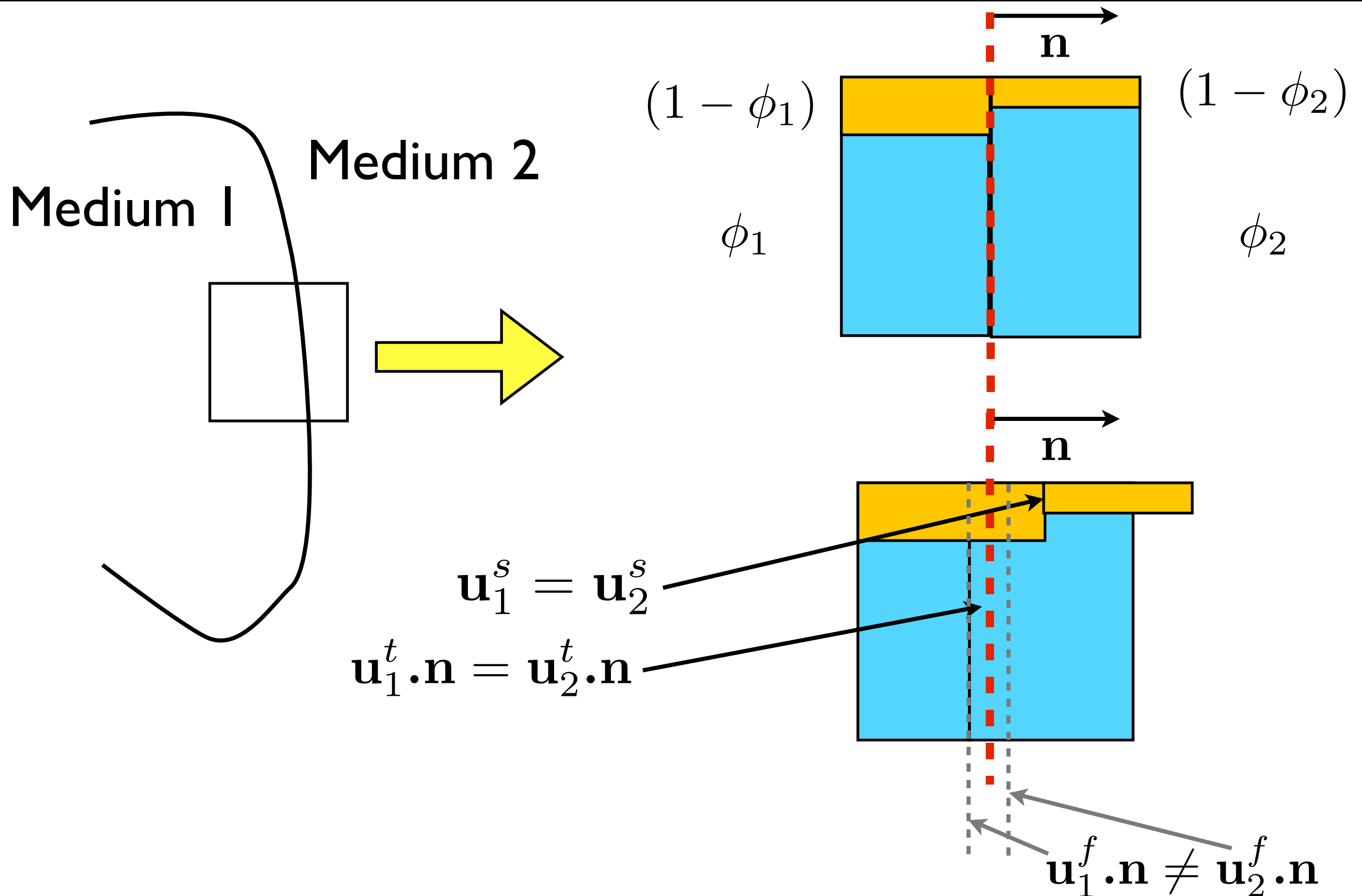
Displacement continuity relations



Displacement continuity relations



Displacement continuity relations



- Displacement
 - Solid in all directions
 - Normal total displacement
 - No conservation of fluid displacement

$$\mathbf{u}_1^s = \mathbf{u}_2^s, \quad \mathbf{u}_1^t \cdot \mathbf{n} = \mathbf{u}_2^t \cdot \mathbf{n}$$

- Stress
 - Conservation of total stress
 - Conservation of pressure
 - No conservation of fluid partial stress

$$\sigma_1^t \cdot \mathbf{n} = \sigma_2^t \cdot \mathbf{n}, \quad p_1 = p_2$$

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$$\sigma_1^t \cdot \mathbf{n} = \sigma_2^t \cdot \mathbf{n}, \quad p_1 = p_2$$

More examples in the following

Viscous effects : Dissipation function

Viscous effects : Dissipation function

Additional terms

$$\mathcal{D} = \frac{\sigma \phi^2 G}{2} \left| \dot{\mathbf{u}}^f - \dot{\mathbf{u}}^s \right|^2$$

$\nearrow \frac{\partial \mathcal{D}}{\partial \dot{\mathbf{u}}^s}$
 $\searrow \frac{\partial \mathcal{D}}{\partial \dot{\mathbf{u}}^f}$

Solid phase

Fluid phase

$$\nabla \cdot \boldsymbol{\sigma}^s = \rho_{11} \frac{\partial^2 \mathbf{u}^s}{\partial t^2} + \rho_{12} \frac{\partial^2 \mathbf{u}^f}{\partial t^2} + \sigma \phi^2 G \left(\frac{\partial \mathbf{u}^s}{\partial t} - \frac{\partial \mathbf{u}^f}{\partial t} \right)$$

$$\nabla \cdot \boldsymbol{\sigma}^f = \rho_{12} \frac{\partial^2 \mathbf{u}^s}{\partial t^2} + \rho_{22} \frac{\partial^2 \mathbf{u}^f}{\partial t^2} + \sigma \phi^2 G \left(\frac{\partial \mathbf{u}^f}{\partial t} - \frac{\partial \mathbf{u}^s}{\partial t} \right)$$

Harmonic excitation

Harmonic excitation

$$\nabla \cdot \boldsymbol{\sigma}^s = -\omega^2 \rho_{11} \mathbf{u}^s - \omega^2 \rho_{12} \mathbf{u}^f + j\omega \sigma \phi^2 G (\mathbf{u}^s - \mathbf{u}^f)$$

$$\nabla \cdot \boldsymbol{\sigma}^f = -\omega^2 \rho_{12} \mathbf{u}^s - \omega^2 \rho_{22} \mathbf{u}^f + j\omega \sigma \phi^2 G (\mathbf{u}^f - \mathbf{u}^s)$$

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$$\nabla \cdot \boldsymbol{\sigma}^s = -\omega^2 \tilde{\rho}_{11} \mathbf{u}^s - \omega^2 \tilde{\rho}_{12} \mathbf{u}^f$$

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Visco-inertial densities

$$\tilde{\rho}_{12} = \rho_{12} - \frac{\sigma \phi^2 G}{j\omega}$$

$$\tilde{\rho}_{22} = \rho_2 - \tilde{\rho}_{12}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}$$

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Link with equivalent fluid models $\mathbf{u}^s = 0$

$$-\phi \nabla p = -\omega^2 \tilde{\rho}_{22} \mathbf{u}^f \implies -\nabla p = -\frac{\omega^2 \tilde{\rho}_{22}}{\phi} \frac{\mathbf{u}^{eq}}{\phi}$$

Harmonic excitation

$$\nabla \cdot \boldsymbol{\sigma}^s = -\omega^2 \tilde{\rho}_{11} \mathbf{u}^s - \omega^2 \tilde{\rho}_{12} \mathbf{u}^f$$

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$$\tilde{\rho}_{22} = \phi^2 \tilde{\rho}_{eq}$$

JKD, Pride

$$A = \frac{(1 - \phi)\left[1 - \phi - \frac{K_b}{K_s}\right]K_s + \phi \frac{K_s K_b}{K_f}}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}} - \frac{2N}{3}$$

$$Q = \frac{\left[1 - \phi - \frac{K_b}{K_s}\right]\phi K_s}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}}$$

$$R = \frac{\phi^2 K_s}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}}$$

Gedanken experiments

$$A = \frac{(1 - \phi)\left[1 - \phi - \frac{K_b}{K_s}\right]K_s + \phi \frac{K_s K_b}{K_f}}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}} - \frac{2N}{3}$$

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$K_b, \quad K_s$ ✓ Structural damping
 ✓ Viscoelasticity
 ✓ AHL

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K_b, K_s ✓ Structural damping
 ✓ Viscoelasticity
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$K_f \longleftarrow \phi \tilde{K}_{eq}$ Champoux-Allard, Lafarge et al



1956 formulation is not valid for non-homogeneous porosity materials

Why ? non continuity of : u^f , σ^f



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Why ? non continuity of : $\mathbf{u}^f$, $\boldsymbol{\sigma}^f$

New representation based on:

$$\mathbf{u}^s, \quad \mathbf{w} = \phi(\mathbf{u}^f - \mathbf{u}^s)$$



M.A. Biot Journal of Applied Physics 33(4) 1962



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$$\mathbf{u}^s, \quad \mathbf{w} = \phi(\mathbf{u}^f - \mathbf{u}^s)$$

Motion equation

$$\begin{aligned} \nabla \cdot \boldsymbol{\sigma}^t(\mathbf{u}^s, \mathbf{w}) &= -\omega^2 \rho_t \mathbf{u}^s - \omega^2 \rho_0 \mathbf{w}, \\ -\nabla p &= -\omega^2 \rho_0 \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{w} \end{aligned}$$



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Constitutive laws

$$\boldsymbol{\sigma}^t(\mathbf{u}^s, \mathbf{w}) = \left[(\hat{A} + \gamma'^2 \tilde{K}_{eq}) \nabla \cdot \mathbf{u}^s + \gamma' \tilde{K}_{eq} \nabla \cdot \mathbf{w} \right] \boldsymbol{\delta} + 2N \boldsymbol{\varepsilon}^s(\mathbf{u}^s)$$

$$p(\mathbf{u}^s, \mathbf{w}) = -\tilde{K}_{eq} [\nabla \cdot \mathbf{w} + \gamma' \nabla \cdot \mathbf{u}^s] \quad \gamma' = 1 - \frac{K_b}{K_s}, \quad \hat{A} = \left(A - \frac{Q^2}{R} \right)$$



M.A. Biot Journal of Applied Physics 33(4) 1962

The Biot theory in practice !

The Biot theory in practice !

$$\Gamma_{3,2i-1} = b_3^+(i) \cos \omega q(i)z \quad (10.133)$$

$$\Gamma_{3,2i} = -jb_3^+(i) \sin \omega q(i)z \quad (10.134)$$

$$M_{1,1} = C_1(1 - r_{11}) - C_2r_{12} + Hr_{13} \quad (8.A.10)$$

$$M_{1,2} = -C_1r_{21} + C_2(1 - r_{22}) + Hr_{23} \quad (8.A.11)$$

$$M_{1,3} = -C_1r_{31} - C_2r_{32} + H(1 + r_{33}) \quad (8.A.12)$$

$$M_{2,1} = -j\alpha_1(1 - r_{11}) + j\alpha_2r_{12} - j\xi r_{13} \quad (8.A.13)$$

$$M_{2,2} = -j\alpha_2(1 - r_{22}) + j\alpha_1r_{21} - j\xi r_{23} \quad (8.A.14)$$

$$M_{2,3} = j\alpha_1r_{31} + j\alpha_2r_{32} - j\xi(1 + r_{33}) \quad (8.A.15)$$

$$M_{3,1} = -j\mu_1\alpha_1(1 - r_{11}) + j\mu_2\alpha_2r_{12} - j\mu_3\xi r_{13} \quad (8.A.16)$$

$$M_{3,2} = -j\mu_2\alpha_2(1 - r_{22}) + j\mu_1\alpha_1r_{21} - j\mu_3\xi r_{23} \quad (8.A.17)$$

$$M_{3,3} = j\mu_1\alpha_1r_{31} + j\mu_2\alpha_2r_{32} - j\mu_3\xi(1 + r_{33}) \quad (8.A.18)$$

$$M_{4,2} = A_2r_{31} + A_2r_{32} - C_3r_{13} \quad (8.A.19)$$

$$M_{4,3} = A_2r_{31} + A_2r_{32} - C_3r_{23} \quad (8.A.20)$$

$$M_{5,1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (8.A.21)$$

$$M_{5,2} = \begin{bmatrix} -(1 - \phi)Z \\ -Z \\ -Z \end{bmatrix} \quad (8.A.22)$$

$$M_{5,3} = \begin{bmatrix} -2Z_1^s \cos \delta_1 l \\ -2Z_1^f \mu_1 \cos \delta_1 l \\ -2Z_2^f \mu_2 \cos \delta_2 l \end{bmatrix} \quad (8.A.23)$$

$$M_{6,1} = \begin{bmatrix} -j\xi(1 + r_{11}) \\ -j\xi(1 + r_{22}) \\ -j\xi(1 + r_{33}) \end{bmatrix} \quad (8.A.24)$$

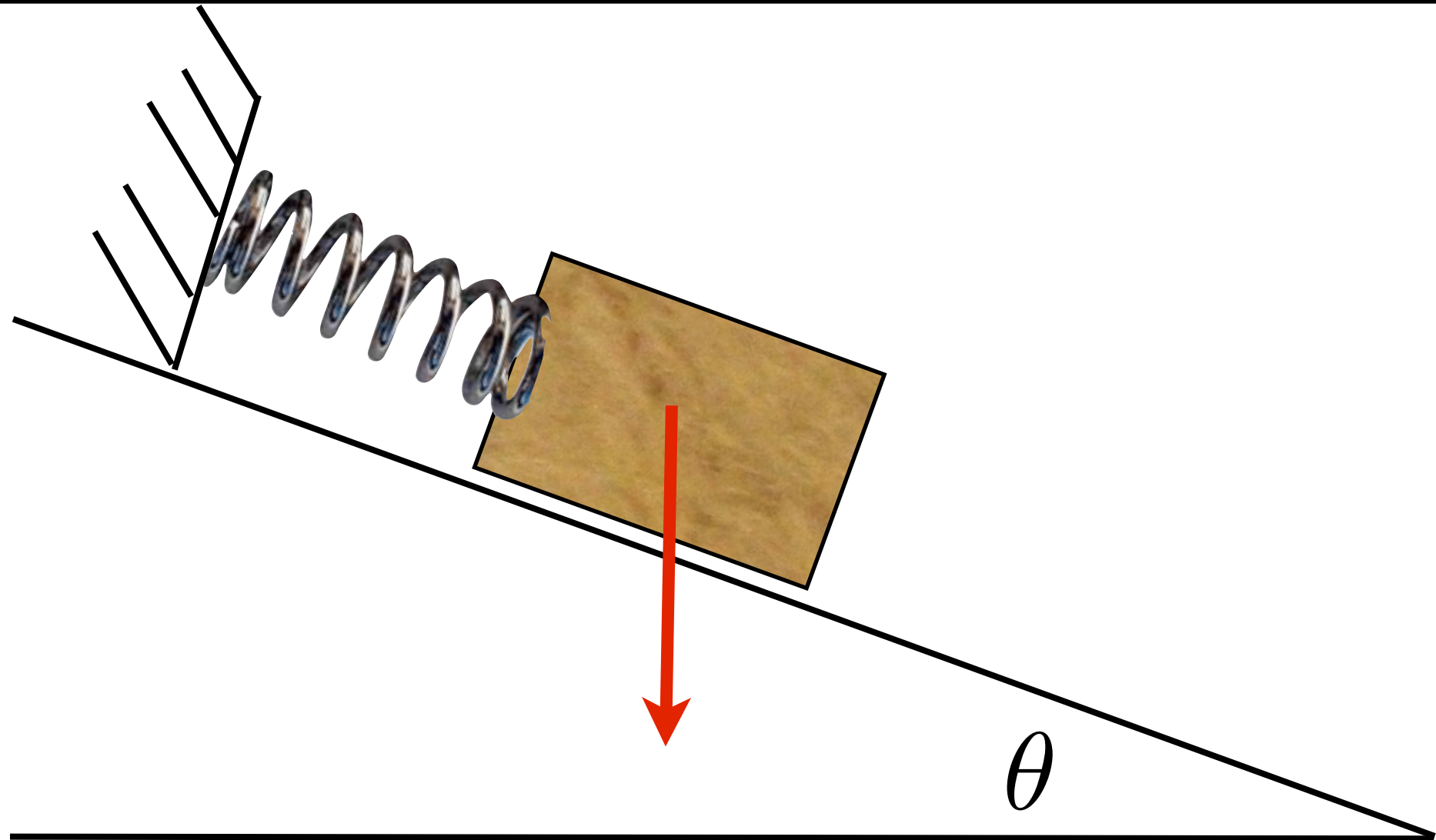
$$M_{6,2} = \begin{bmatrix} -j\xi r_{12} \\ -j\xi r_{21} \\ -j\xi r_{31} \end{bmatrix} \quad (8.A.25)$$

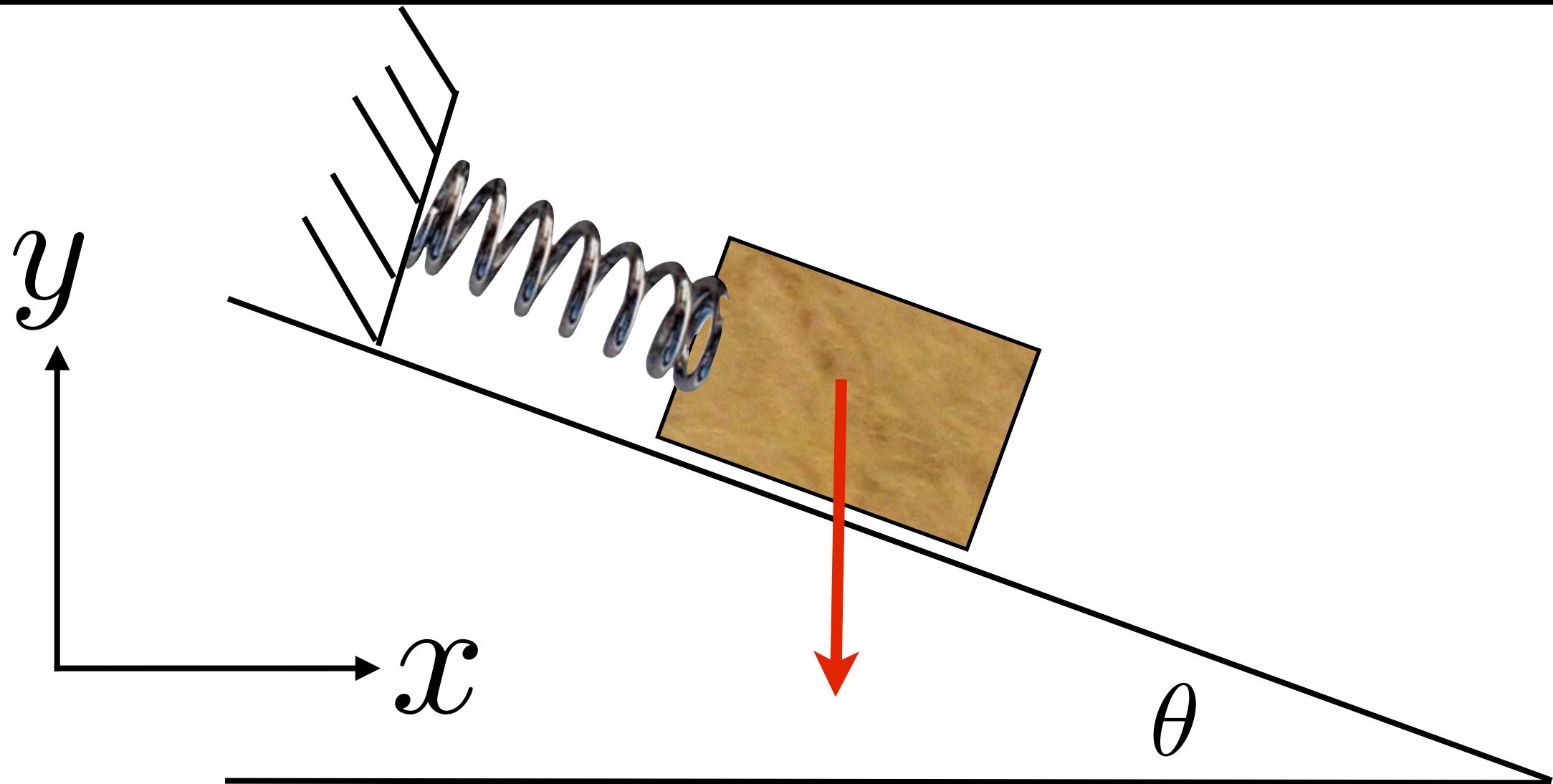
$$M_{6,3} = j\alpha_3(1 - r_{33}) - j\xi r_{31} \quad (8.A.26)$$

$$\delta_1^2 = \frac{\omega^2}{2(PR - Q^2)} [P\tilde{\rho}_{22} + R\tilde{\rho}_{11} - 2Q\tilde{\rho}_{12} - \sqrt{\Delta}] \quad (10.140)$$

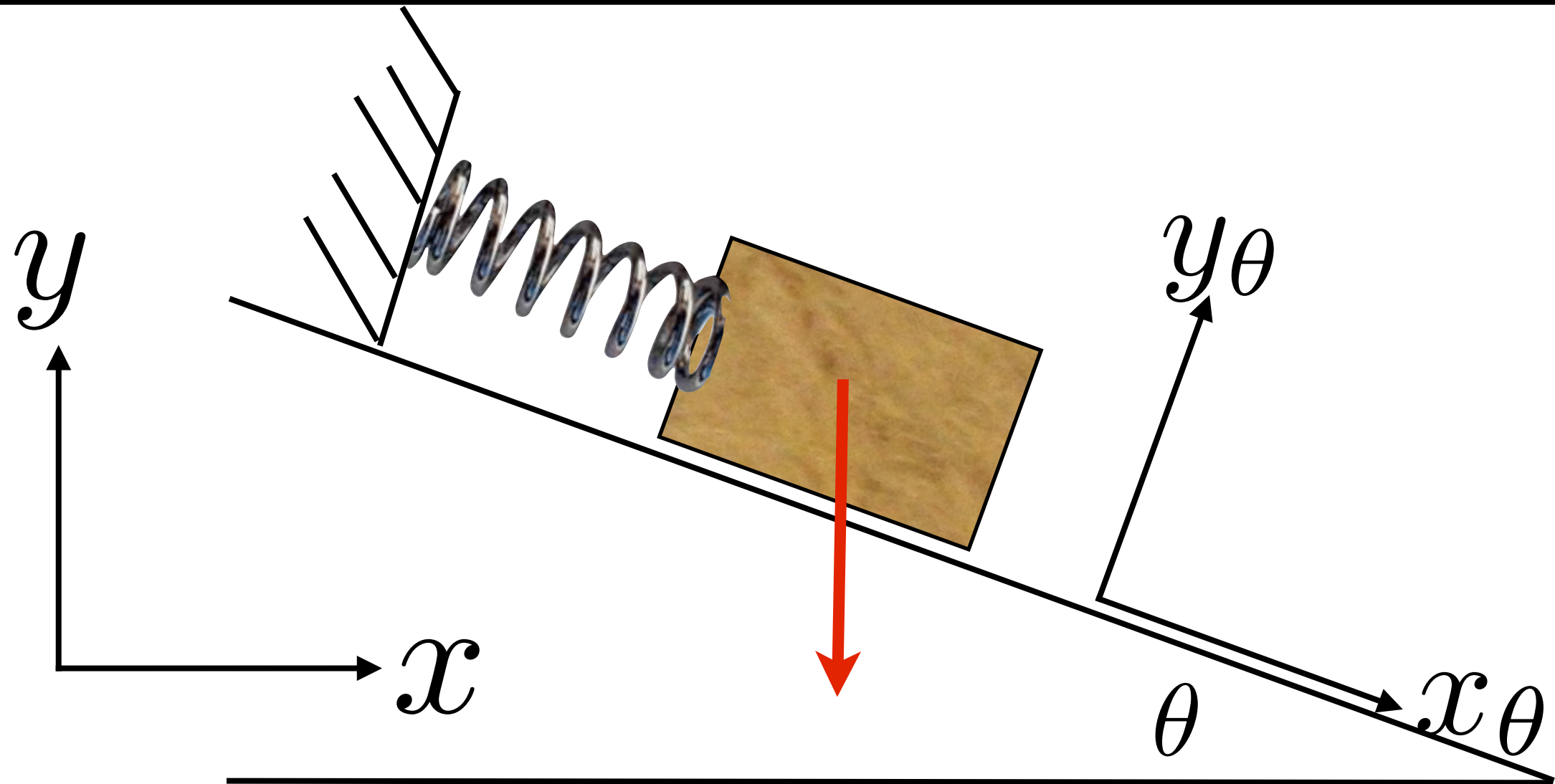
$$\delta_2^2 = \frac{\omega^2}{2(PR - Q^2)} [P\tilde{\rho}_{22} + R\tilde{\rho}_{11} - 2Q\tilde{\rho}_{12} - \sqrt{\Delta}] \quad (10.141)$$

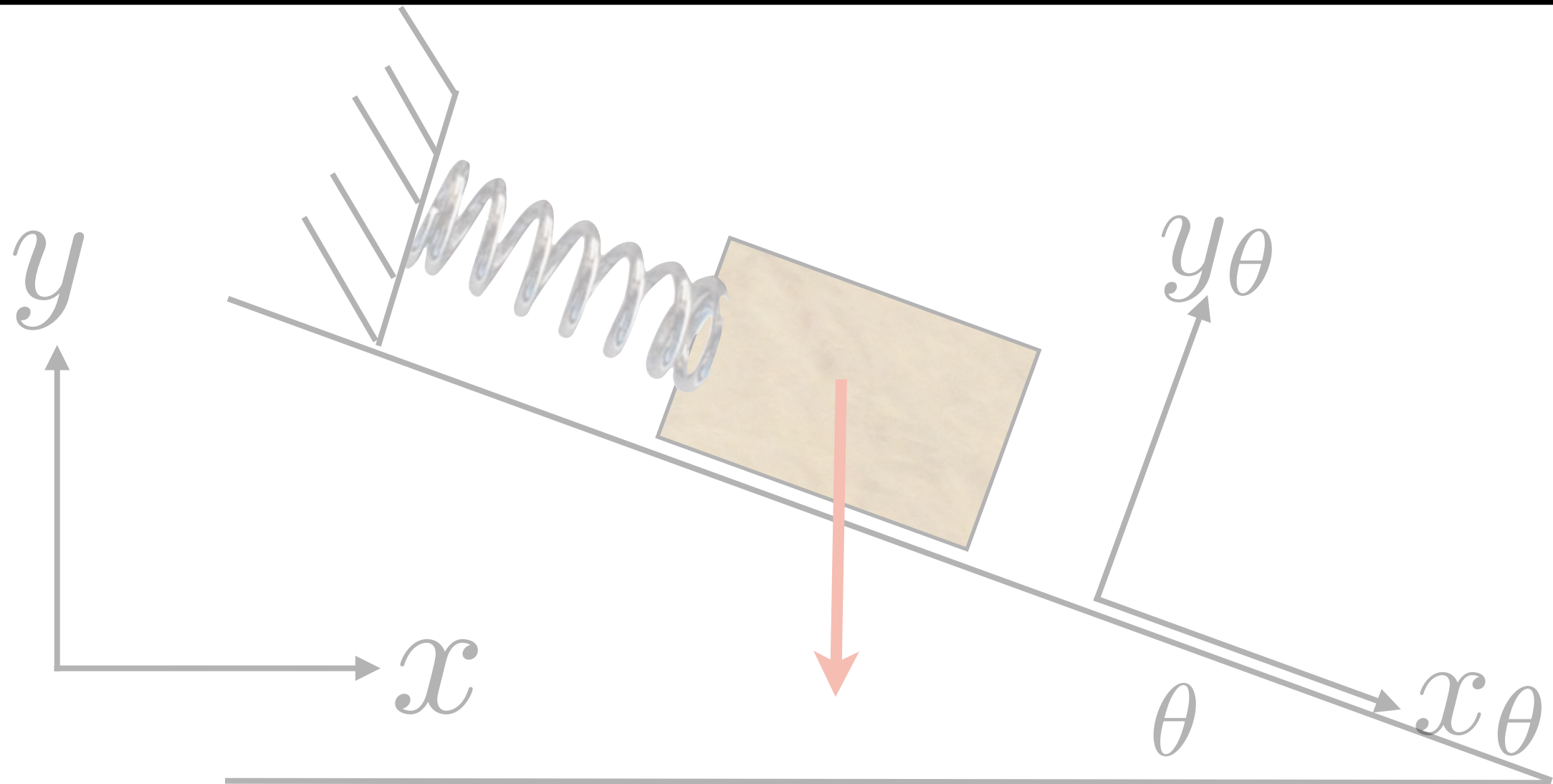
Pencil consuming, Maple addict...



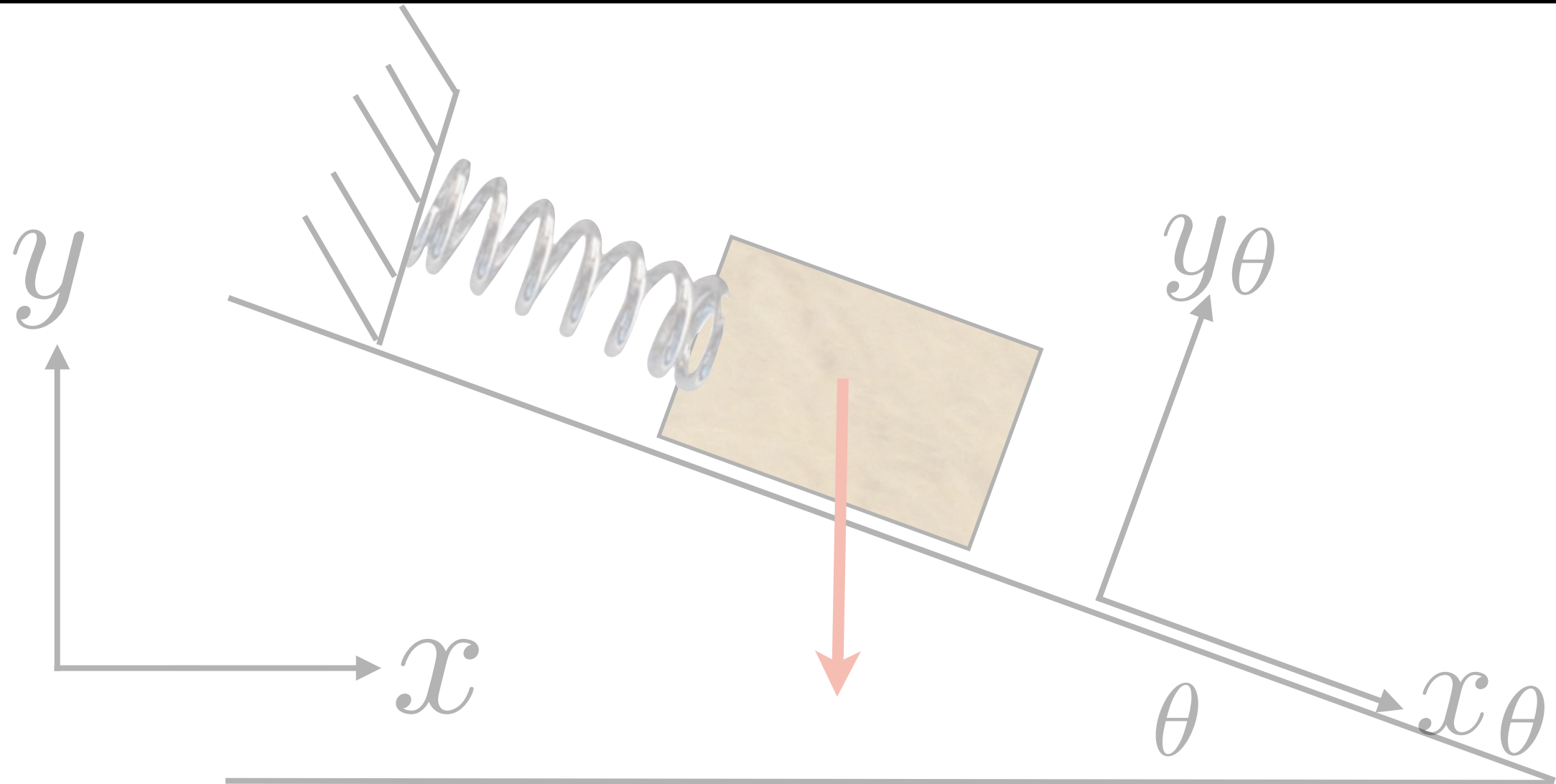


Toward simplifications





$$f(x) + y = y$$



$$f(x) + y = y \quad \Longleftrightarrow \quad f(x) = 0$$

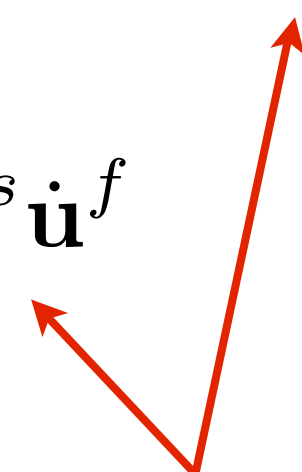
In 1956 and 1962 formulations, fields are imbricated

In 1956 and 1962 formulations, fields are imbricated

- Motion equations

$$\mathcal{W}_{56} = \tilde{A} \frac{\nabla \cdot \mathbf{u}^s \nabla \cdot \mathbf{u}^s}{2} + 2N \varepsilon_{ij}^s \varepsilon_{ij}^s + \tilde{R} \frac{\nabla \cdot \mathbf{u}^f \nabla \cdot \mathbf{u}^f}{2} + \tilde{Q} \nabla \cdot \mathbf{u}^s \nabla \cdot \mathbf{u}^f$$

$$\mathcal{T}_{56} = \tilde{\rho}_{11} \frac{\dot{\mathbf{u}}^s \dot{\mathbf{u}}^s}{2} + \tilde{\rho}_{22} \frac{\dot{\mathbf{u}}^f \dot{\mathbf{u}}^f}{2} + \tilde{\rho}_{12} \dot{\mathbf{u}}^s \dot{\mathbf{u}}^f$$



Coupling terms

In 1956 and 1962 formulations, fields are imbricated

- Motion equations
- Constitutive laws

$$\sigma_{ij}^s(\mathbf{u}^s, \mathbf{u}^f) = \tilde{A} \nabla \cdot \mathbf{u}^s \delta_{ij} + 2N \varepsilon_{ij} + \tilde{Q} \nabla \cdot \mathbf{u}^f \delta_{ij}$$

In 1956 and 1962 formulations, fields are imbricated

- Motion equations
- Constitutive laws
- Parameters

$$\tilde{A} = \frac{(1 - \phi) \left[1 - \phi - \frac{K_b}{K_s} \right] K_s + \phi \frac{K_s K_b}{K_f}}{1 - \phi - \frac{K_b}{K_s} + \phi \frac{K_s}{K_f}} - \frac{2N}{3}$$

In 1956 and 1962 formulations, fields are imbricated

- Motion equations
- Constitutive laws
- Parameters
- Conservation laws

$$(1 - \phi_1)\mathbf{u}_1^s + \phi_1\mathbf{u}_1^f = (1 - \phi_2)\mathbf{u}_2^s + \phi_1\mathbf{u}_2^f$$

In 1956 and 1962 formulations, fields are imbricated

- Motion equations
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$$(1 - \phi_1)\mathbf{u}_1^s + \phi_1\mathbf{u}_1^f = (1 - \phi_2)\mathbf{u}_2^s + \phi_1\mathbf{u}_2^f$$

Idea: Diagonalization (rotation) of strain energy density

$$\begin{Bmatrix} \mathbf{u}^s \\ \mathbf{u}^f \end{Bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix}, \quad ad - bc \neq 0$$

$$\begin{Bmatrix} \mathbf{u}^s \\ \mathbf{u}^f \end{Bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix}, \quad ad - bc \neq 0$$

$$\begin{aligned} \mathcal{V} = & \frac{(\nabla \cdot \mathbf{u}_1)^2}{2} (Aa^2 + Rc^2 + 2Qac) \\ & + \frac{(\nabla \cdot \mathbf{u}_2)^2}{2} (Ab^2 + Rd^2 + 2Qbd) \\ & + \nabla \cdot \mathbf{u}_1 \nabla \cdot \mathbf{u}_2 (Aab + Rcd + Q(ad + bc)) \\ & + 2N (a^2 \varepsilon_{ij}^1 \varepsilon_{ij}^1 + b^2 \varepsilon_{ij}^2 \varepsilon_{ij}^2 + 2ab \varepsilon_{ij}^1 \varepsilon_{ij}^2) \end{aligned}$$

$$\begin{Bmatrix} \mathbf{u}^s \\ \mathbf{u}^f \end{Bmatrix} = \begin{bmatrix} a & \cancel{b} \\ c & d \end{bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix}, \quad ad - bc \neq 0$$

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$$\begin{Bmatrix} \mathbf{u}^s \\ \mathbf{u}^f \end{Bmatrix} = \begin{bmatrix} a & \cancel{b} \\ c & d \end{bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix}, \quad ad - bc \neq 0$$

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$\times b = 0$

$\rightarrow c = -\frac{Q}{R} a$

$$\begin{Bmatrix} \mathbf{u}^s \\ \mathbf{u}^f \end{Bmatrix} = \begin{bmatrix} a & \cancel{b} \\ c & d \end{bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix}, \quad ad - bc \neq 0$$

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$\times b = 0$
 $c = -\frac{Q}{R} a$
 $a = 1$

$$\begin{Bmatrix} \mathbf{u}^s \\ \mathbf{u}^f \end{Bmatrix} = \begin{bmatrix} a & \cancel{b} \\ c & d \end{bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix}, \quad ad - bc \neq 0$$

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$\times b = 0$
 $c = -\frac{Q}{R} a$
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 $d = \frac{1}{\phi}$

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 $d = \frac{1}{\phi}$

Fields of representation

$$\mathbf{u}^s \quad \text{and} \quad \mathbf{u}^W = \phi \left(\mathbf{u}^f + \frac{Q}{R} \mathbf{u}^s \right) = \mathbf{u}^t - \frac{K_b}{K_s} \mathbf{u}^s$$

$$\mathcal{T} = \frac{\rho_s}{2} \dot{\mathbf{u}}^s{}^2 + \frac{\rho_{eq}}{2} \dot{\mathbf{u}}^W{}^2 + \rho_{eq} \gamma \dot{\mathbf{u}}^s \dot{\mathbf{u}}^W$$

$$\mathcal{V} = \hat{A} \frac{(\nabla \cdot \mathbf{u}^s)^2}{2} + 2N \varepsilon_{ij}^s \varepsilon_{ij}^s + \frac{R}{\phi^2} \frac{(\nabla \cdot \mathbf{u}^W)^2}{2}$$

$$\mathcal{V} = \hat{A} \frac{(\nabla \cdot \mathbf{u}^s)^2}{2} + 2N \varepsilon_{ij}^s \varepsilon_{ij}^s + \frac{R}{\phi^2} \frac{(\nabla \cdot \mathbf{u}^W)^2}{2}$$

Motion equations (with complex coefficients)

$$\begin{aligned} \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{u}^s) &= -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^W, \\ -\nabla p(\mathbf{u}^W) &= -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{u}^W \end{aligned}$$

$$\hat{\sigma}_{ij}(\mathbf{u}^s) = \hat{A} \nabla \cdot \mathbf{u}^s \delta_{ij} + 2N \varepsilon_{ij}^s \quad p(\mathbf{u}^W) = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^W$$

Dazel et al. JASA 121(6) 2007

$$\tilde{\gamma} = \phi \left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}} \right) \quad \tilde{\rho}_s = \underbrace{\tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}^2}{\tilde{\rho}_{22}}}_{\tilde{\rho}} + \tilde{\gamma}^2 \tilde{\rho}_{eq}$$

Total displacement

$$\mathbf{u}^W = \mathbf{u}^t$$

$$\hat{\sigma}_{ij} = \sigma_{ij}^t + p \delta_{ij}$$

$$\mathbf{u}^s, \quad \mathbf{u}^t \cdot \mathbf{n}$$

Total displacement

$$\mathbf{u}^W = \mathbf{u}^t$$

Porous-porous interface

$$\hat{\sigma}_{ij} = \sigma_{ij}^t + p \delta_{ij}$$

$$\mathbf{u}^s, \mathbf{u}^t \cdot \mathbf{n}$$

Conservation at interface between 2 PEM

Total displacement

$$\mathbf{u}^W = \mathbf{u}^t$$

Porous-porous interface

$$\hat{\sigma}_{ij} = \sigma_{ij}^t + p \delta_{ij}$$

$$\mathbf{u}^s, \mathbf{u}^t \cdot \mathbf{n}$$

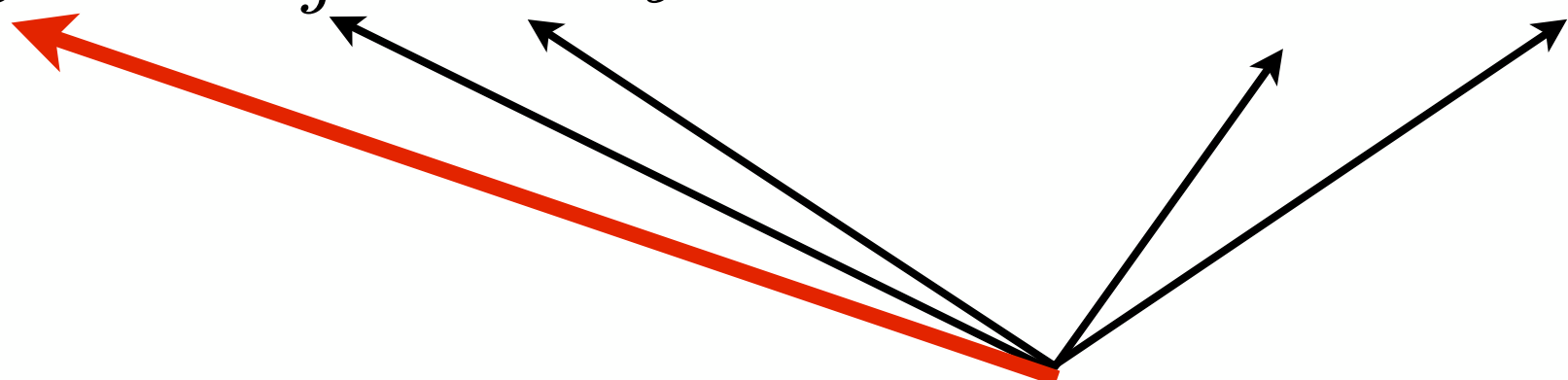
Conservation at interface between 2 PEM

Total displacement

$$\mathbf{u}^W = \mathbf{u}^t$$

Porous-porous interface

$$\hat{\sigma}_{ij} = \sigma_{ij}^t + p \delta_{ij}$$

$$\mathbf{u}^s, \mathbf{u}^t \cdot \mathbf{n}$$


Conservation at interface between 2 PEM

Porous-air interface

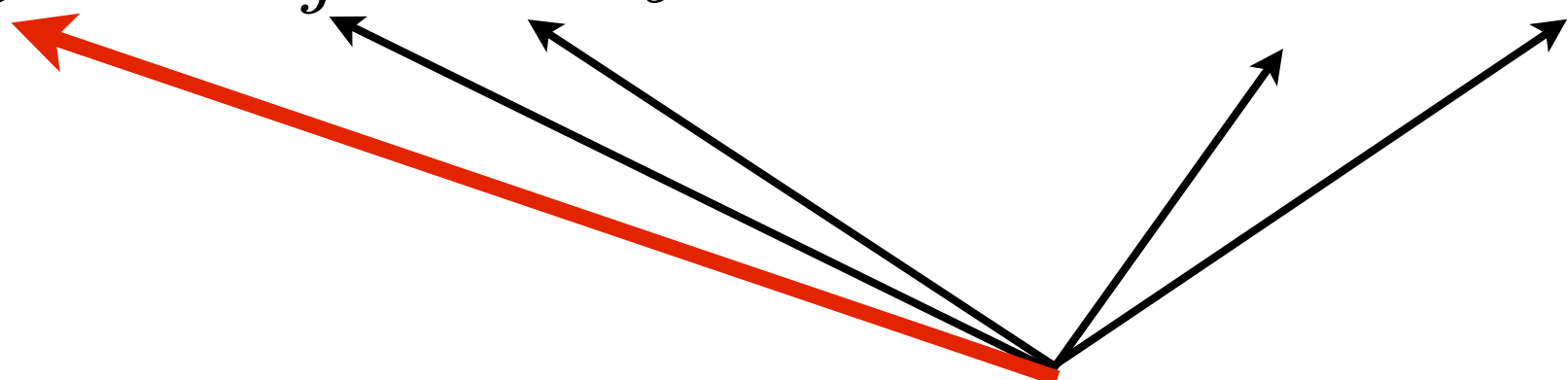
$$p^a = p, \quad \hat{\sigma} \cdot \mathbf{n} = 0 \qquad \mathbf{u}^a \cdot \mathbf{n} = \mathbf{u}^t \cdot \mathbf{n}$$

Total displacement

$$\mathbf{u}^W = \mathbf{u}^t$$

Porous-porous interface

$$\hat{\sigma}_{ij} = \sigma_{ij}^t + p \delta_{ij}$$

$$\mathbf{u}^s, \mathbf{u}^t \cdot \mathbf{n}$$


Conservation at interface between 2 PEM

Porous-air interface

$$p^a = p, \quad \hat{\sigma} \cdot \mathbf{n} = 0 \quad \mathbf{u}^a \cdot \mathbf{n} = \mathbf{u}^t \cdot \mathbf{n}$$

Natural coupling

$$\tilde{A} = \frac{(1 - \phi)[1 - \phi - \frac{K_b}{K_s}]K_s + \frac{K_s K_b}{\tilde{K}_{eq}}}{1 - \phi - \frac{K_b}{K_s} + \frac{K_s}{\tilde{K}_{eq}}} - \frac{2N}{3}$$

$$\tilde{Q} = \frac{[1 - \phi - \frac{K_b}{K_s}]\phi K_s}{1 - \phi - \frac{K_b}{K_s} + \frac{K_s}{\tilde{K}_{eq}}}$$

$$\tilde{R} = \frac{\phi^2 K_s}{1 - \phi - \frac{K_b}{K_s} + \frac{K_s}{\tilde{K}_{eq}}}$$

$$\tilde{A} = \frac{(1 - \phi)[1 - \phi - \cancel{\frac{K_b}{K_s}}]K_s + \frac{K_s K_b}{\tilde{K}_{eq}}}{1 - \phi - \cancel{\frac{K_b}{K_s}} + \frac{K_s}{\tilde{K}_{eq}}} - \frac{2N}{3}$$

$$\tilde{Q} = \frac{[1 - \phi - \cancel{\frac{K_b}{K_s}}]\phi K_s}{1 - \phi - \cancel{\frac{K_b}{K_s}} + \frac{K_s}{\tilde{K}_{eq}}}$$

$$\tilde{R} = \frac{\phi^2 K_s}{1 - \phi - \cancel{\frac{K_b}{K_s}} + \frac{K_s}{\tilde{K}_{eq}}}$$

$$\cancel{X} K_b \ll K_s$$

$$\tilde{A} = \frac{(1 - \phi)[1 - \phi - \frac{K_b}{K_s}]K_s + \frac{K_s K_b}{\tilde{K}_{eq}}}{\cancel{1 - \phi - \frac{K_b}{K_s}} + \frac{K_s}{\tilde{K}_{eq}}} - \frac{2N}{3}$$

$$\tilde{Q} = \frac{[1 - \phi - \frac{K_b}{K_s}]\phi K_s}{\cancel{1 - \phi - \frac{K_b}{K_s}} + \frac{K_s}{\tilde{K}_{eq}}}$$

$$\tilde{R} = \frac{\phi^2 K_s}{\cancel{1 - \phi - \frac{K_b}{K_s}} + \frac{K_s}{\tilde{K}_{eq}}}$$

$$\begin{aligned} & \times K_b \ll K_s \\ & \times \tilde{K}_{eq} \ll K_s \end{aligned}$$

$$\tilde{A} = \underbrace{K_b - \frac{2N}{3}}_{\hat{A}} + (1 - \phi)^2 \tilde{K}_{eq}$$

$$\tilde{Q} = (1 - \phi)\phi \tilde{K}_{eq}$$

$$\tilde{R} = \phi^2 \tilde{K}_{eq}$$

$$\left. \begin{aligned} \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{u}^s) &= -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^t, \\ -\nabla p(\mathbf{u}^t) &= -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{u}^t \end{aligned} \right| \quad p = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^t$$

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$$p = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^t$$

$$\mathbf{u}^t = \frac{1}{\omega^2 \tilde{\rho}_{eq}} \nabla p - \tilde{\gamma} \mathbf{u}^s$$

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$$\mathbf{u}^t = \frac{1}{\omega^2 \tilde{\rho}_{eq}} \nabla p - \tilde{\gamma} \mathbf{u}^s$$

Motion equation of the solid (vectorial)

$$\nabla \cdot \hat{\sigma}(\mathbf{u}^s) = -\omega^2 \tilde{\rho} \mathbf{u}^s - \tilde{\gamma} \nabla p$$

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$$\left. \begin{aligned} \nabla \cdot \hat{\sigma}(\mathbf{u}^s) &= -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^t, \\ \nabla \cdot - \nabla p(\mathbf{u}^t) &= -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{u}^t \end{aligned} \right| \quad p = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^t$$
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Motion equation of pressure (scalar)

Atalla et al. JASA 104 1998

$$-\nabla^2 p(\mathbf{u}^t) = -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \nabla \cdot \mathbf{u}^s + \omega^2 \frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}} p$$

$$\left. \begin{aligned} \nabla \cdot \hat{\sigma}(\mathbf{u}^s) &= -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^t, \\ \nabla \cdot -\nabla p(\mathbf{u}^t) &= -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{u}^t \end{aligned} \right| \quad p = -\tilde{K}_{eq} \nabla \cdot \mathbf{u}^t$$

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Atalla et al. JASA 104 1998

$$-\nabla^2 p(\mathbf{u}^t) = -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \nabla \cdot \mathbf{u}^s + \omega^2 \frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}} p$$

Only 4 equations are necessary from a dynamical point of view



Cotton



Mineral wools

Limp



Metallic foams

Light rigid



Cotton



Mineral wools



Metallic foams

Limp

Light rigid

Strain energy of the solid negligible compared to kinetic energy



Cotton



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Strain energy of the solid negligible compared to kinetic energy

$$\cancel{\nabla \cdot \hat{\sigma}(\mathbf{u}^s)} = -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^W,$$
$$-\nabla p(\mathbf{u}^W) = -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{u}^W$$



Cotton



Mineral wools



Metallic foams

Limp

Light rigid

Strain energy of the solid negligible compared to kinetic energy

$$\begin{aligned}
 \cancel{\nabla \cdot \hat{\sigma}(\mathbf{u}^s)} &= -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^W, & \mathbf{u}^s &= -\frac{\tilde{\gamma} \tilde{\rho}_{eq}}{\tilde{\rho}_s} \mathbf{u}^t \\
 -\nabla p(\mathbf{u}^W) &= -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \mathbf{u}^W
 \end{aligned}$$



Cotton



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Metallic foams

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Strain energy of the solid negligible compared to kinetic energy

$$\cancel{\nabla \cdot \hat{\sigma}(\mathbf{u}^s)} = -\omega^2 \tilde{\rho}_s \mathbf{u}^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^W,$$

$$-\nabla p(\mathbf{u}^W) = -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} \mathbf{u}^s \leftarrow -\omega^2 \tilde{\rho}_{eq} \mathbf{u}^W \quad \mathbf{u}^s = -\frac{\tilde{\gamma} \tilde{\rho}_{eq}}{\tilde{\rho}_s} \mathbf{u}^t$$

$$-\nabla p = -\omega^2 \tilde{\rho}_l \mathbf{u}^t \implies \nabla^2 p + \delta_{limp}^2 p = 0$$

$$\tilde{\rho}_l = \tilde{\rho}_{eq} \frac{\tilde{\rho}}{\tilde{\rho}_s} \quad \delta_{limp} = \omega \sqrt{\frac{\tilde{\rho}_l}{\tilde{K}_{eq}}} \quad Z_{limp} = \sqrt{\tilde{\rho}_l \tilde{K}_{eq}}$$

General form of poroelastic plane waves

$$\left\{ \frac{u^s}{u^t} \right\} = \left\{ \begin{array}{c} u_1^s \\ u_2^s \\ u_3^s \\ \frac{u_3^s}{u_1^t} \\ u_1^t \\ u_2^t \\ u_3^t \end{array} \right\} e^{j(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

**General form of poroelastic
plane waves**

$$\left\{ \frac{u^s}{u^t} \right\} = \left\{ \begin{array}{c} u_1^s \\ u_2^s \\ u_3^s \\ \hline u_1^t \\ u_2^t \\ u_3^t \end{array} \right\} e^{j(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

Wave number

Isotropic materials

$\mathbf{k} = \delta \mathbf{e}(\theta, \phi)$

**General form of poroelastic
plane waves**

$$\left\{ \frac{u^s}{u^t} \right\} = \left\{ \begin{array}{c} u_1^s \\ u_2^s \\ u_3^s \\ \frac{u_1^t}{u_2^t} \\ u_3^t \end{array} \right\} e^{j(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

Wave number $\swarrow$

Polarisation $\nearrow$
(Wave in both phases)

Isotropic materials
 $\mathbf{k} = \delta \mathbf{e}(\theta, \phi)$

General form of poroelastic plane waves

$$\left\{ \frac{u^s}{u^t} \right\} = \left\{ \begin{array}{c} u_1^s \\ u_2^s \\ u_3^s \\ u_1^t \\ u_2^t \\ u_3^t \end{array} \right\} e^{j(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

Wave number $\mathbf{k} = \delta \mathbf{e}(\theta, \phi)$

Isotropic materials

Polarisation
(Wave in both phases)

✓ Approach 1: direct resolution (valid for anisotropic materials)

✓ Approach 2: Potentials

✓ Compressional waves $\mathbf{u}^s = \nabla \varphi^s, \quad \mathbf{u}^t = \nabla \varphi^t$

✓ Rotational waves $\mathbf{u}^s = \nabla \wedge \psi^s, \quad \mathbf{u}^t = \nabla \wedge \psi^t$

Diagonalization of the problem

$$-\omega^2 \begin{bmatrix} \tilde{\rho}_s & \tilde{\rho}_{eq}\tilde{\gamma} \\ \tilde{\rho}_{eq}\tilde{\gamma} & \tilde{\rho}_{eq} \end{bmatrix} \begin{Bmatrix} \varphi^s \\ \varphi^t \end{Bmatrix} = \begin{bmatrix} \hat{P} & 0 \\ 0 & \tilde{K}_{eq} \end{bmatrix} \nabla^2 \begin{Bmatrix} \varphi^s \\ \varphi^t \end{Bmatrix}$$

Diagonalization of the problem

$$\delta_{1,2} = \frac{(\delta_{s2}^2 + \delta_{eq}^2) \pm \sqrt{(\delta_{s2}^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2\delta_{s1}^2}}{2}$$

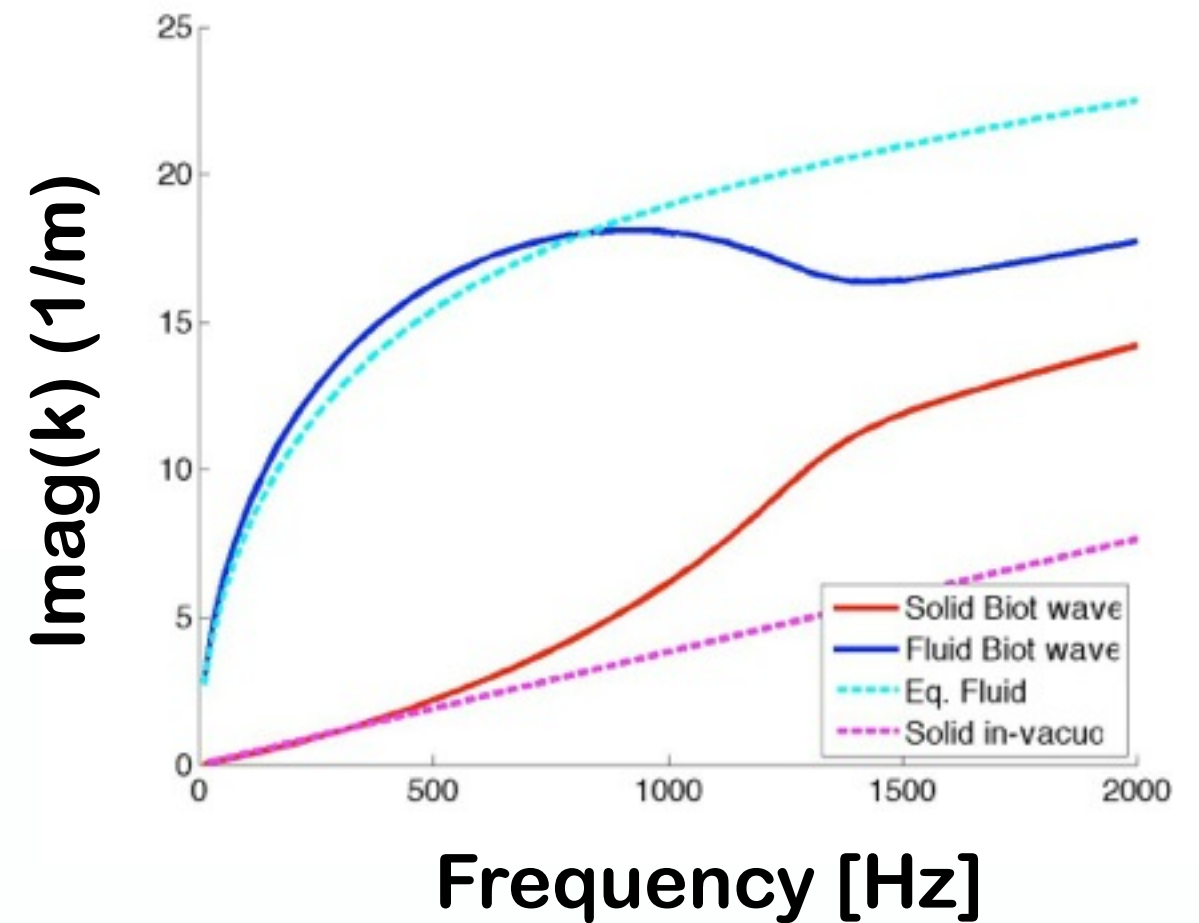
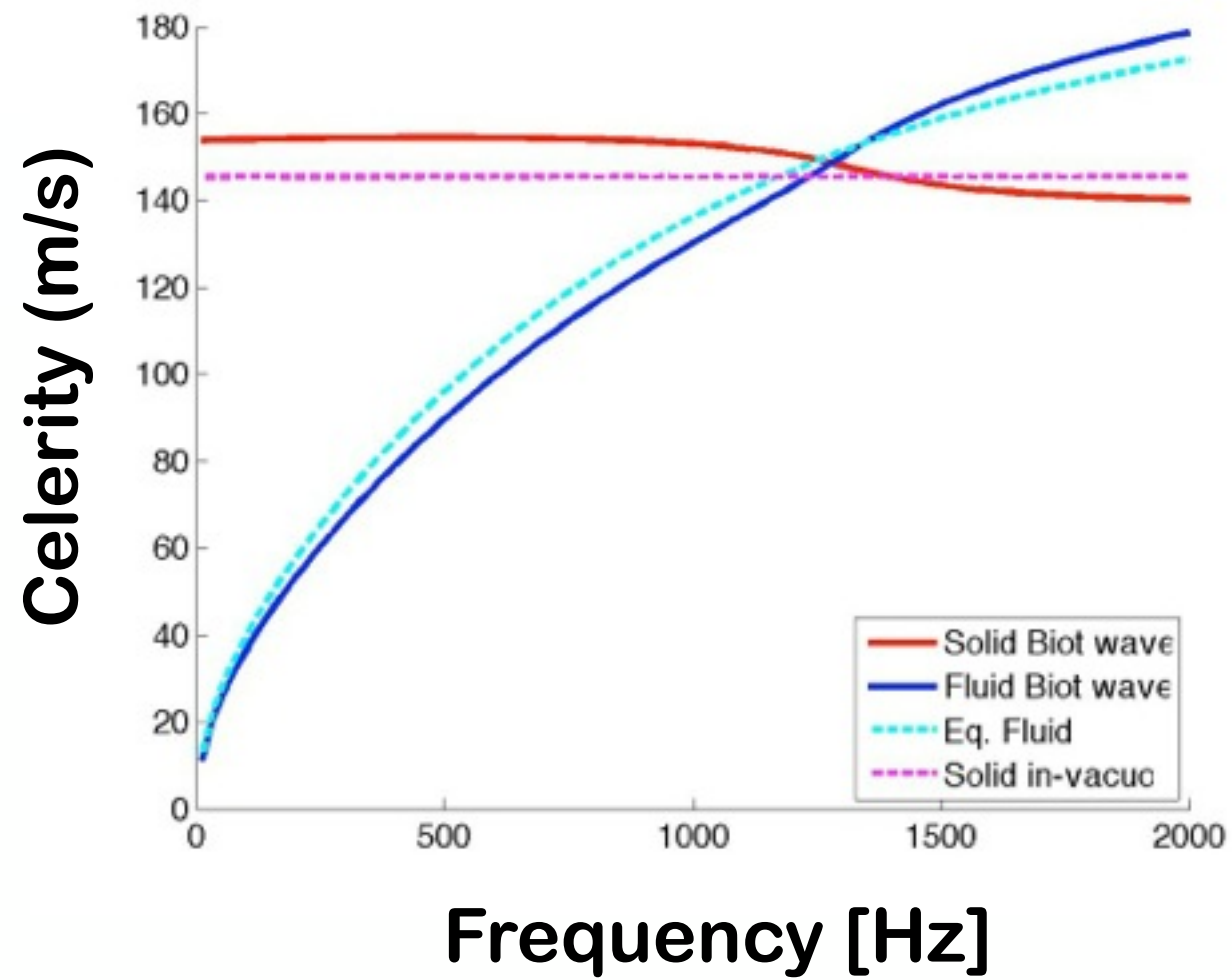
Diagonalization of the problem

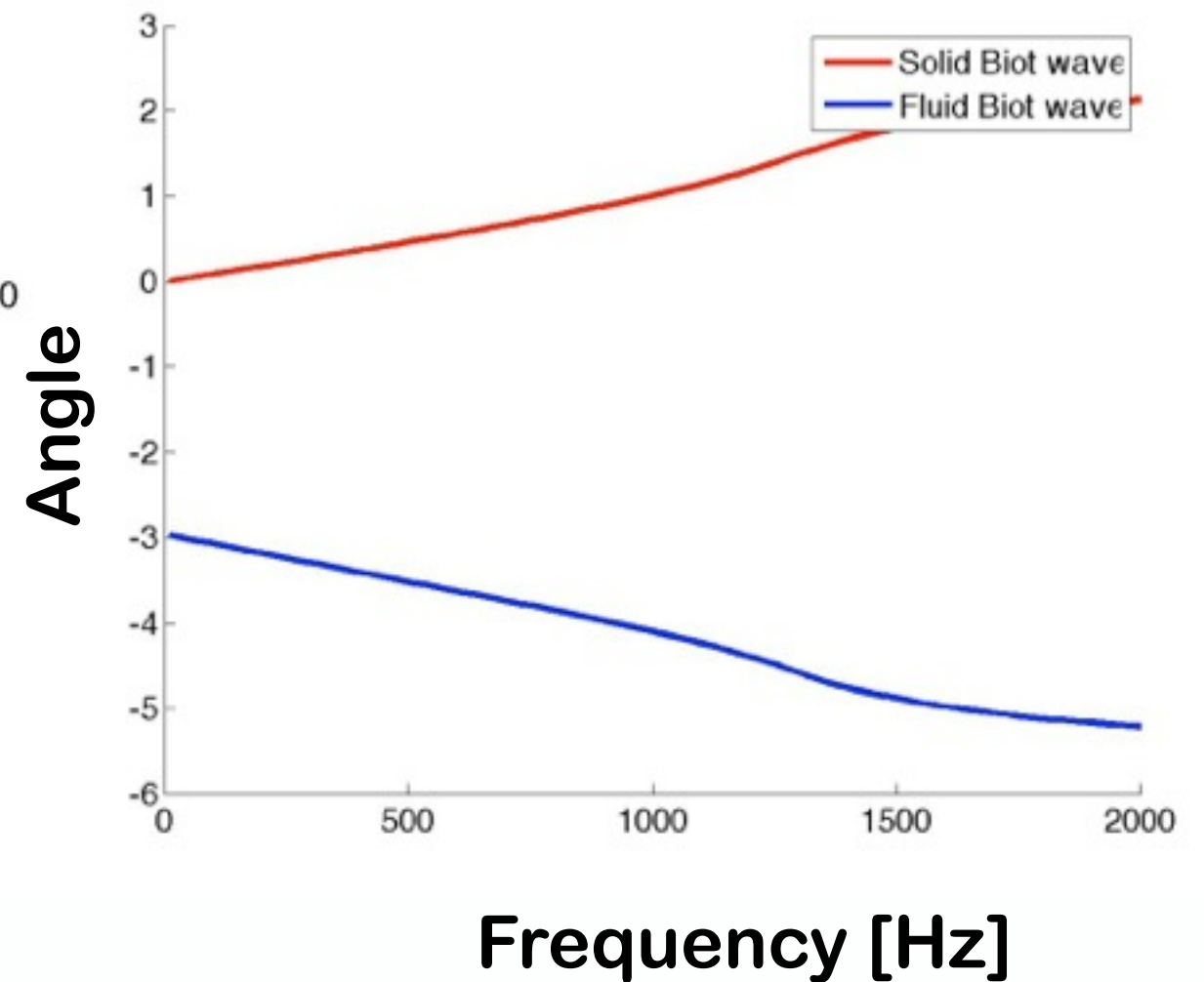
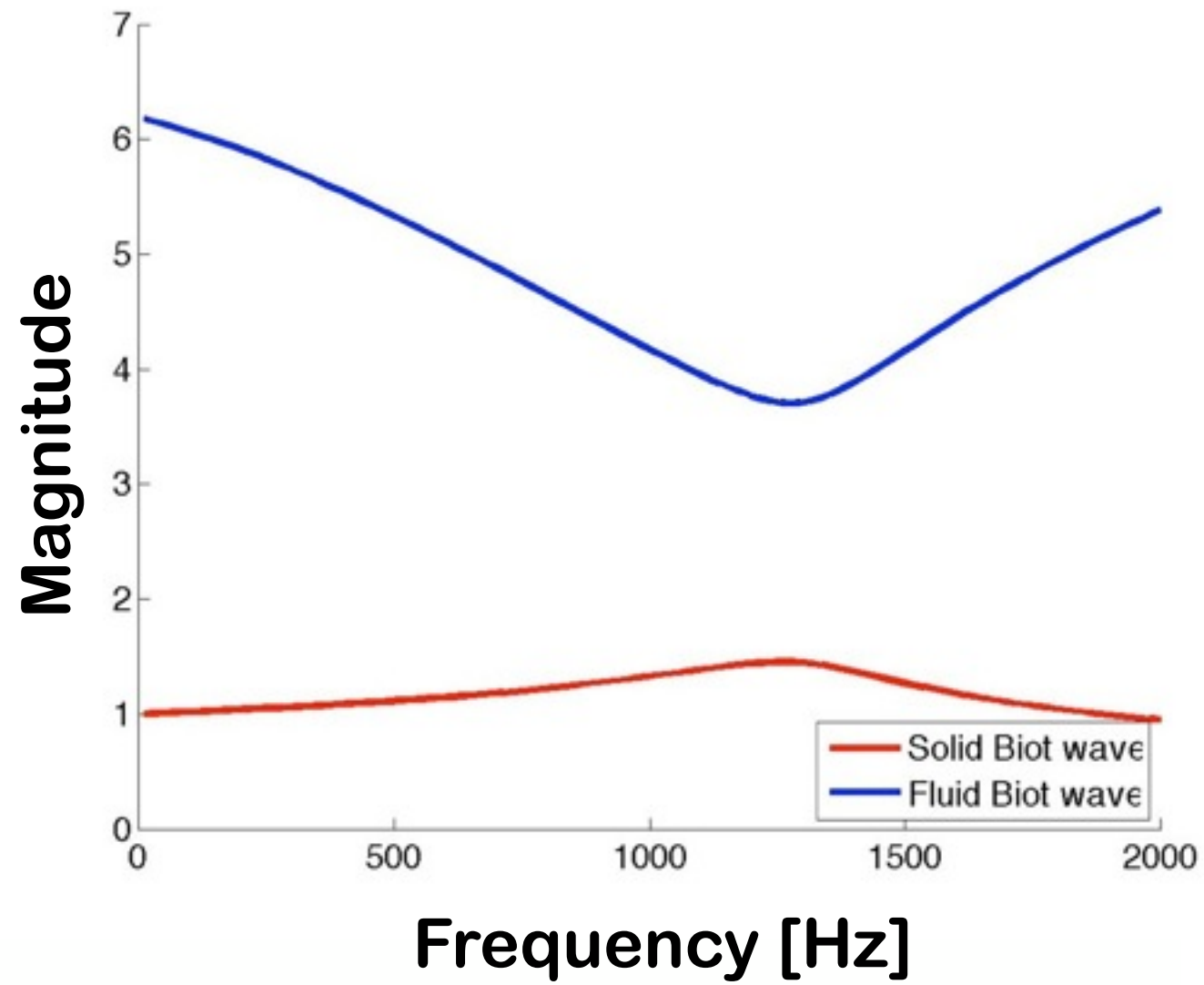
Intrinsic wave numbers

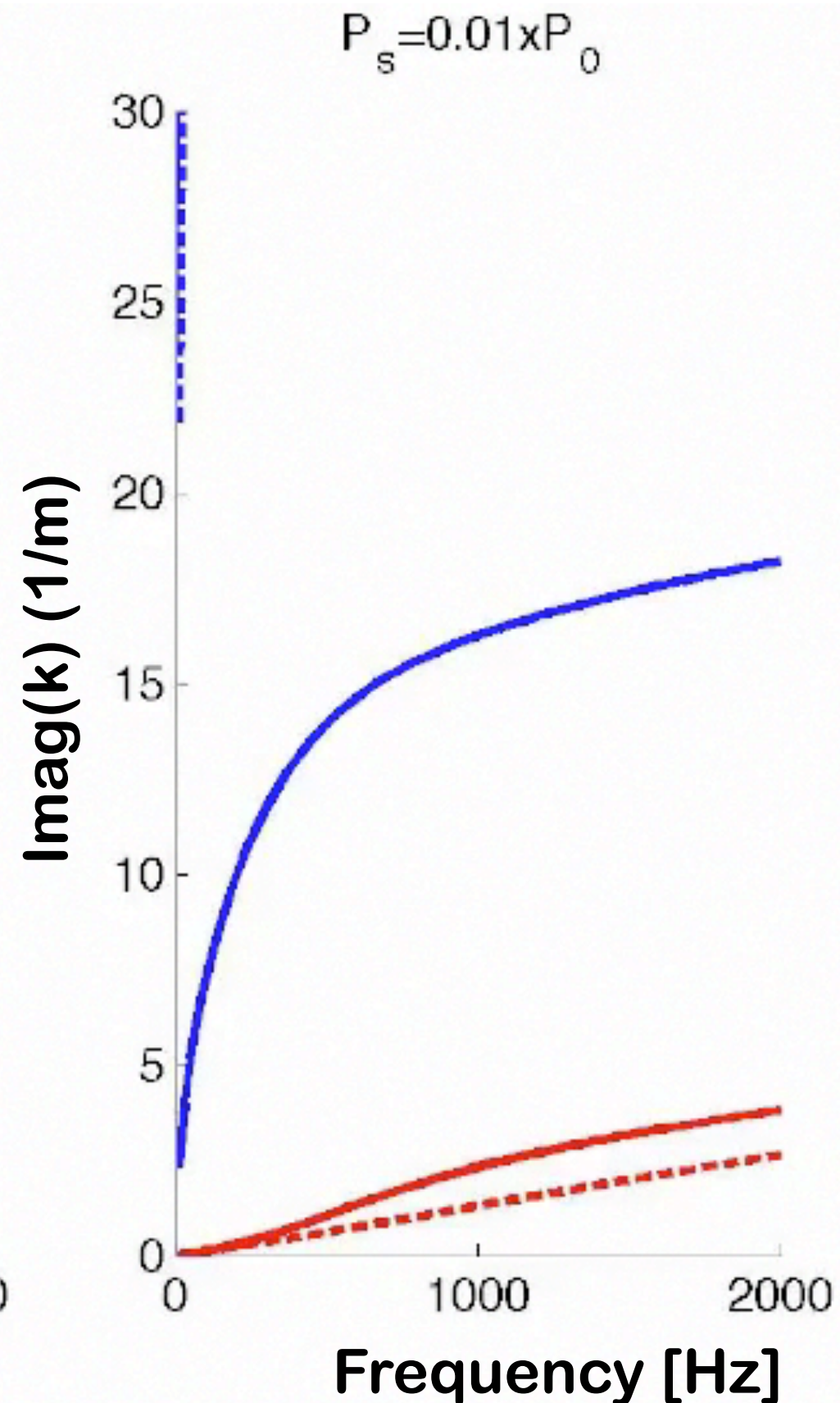
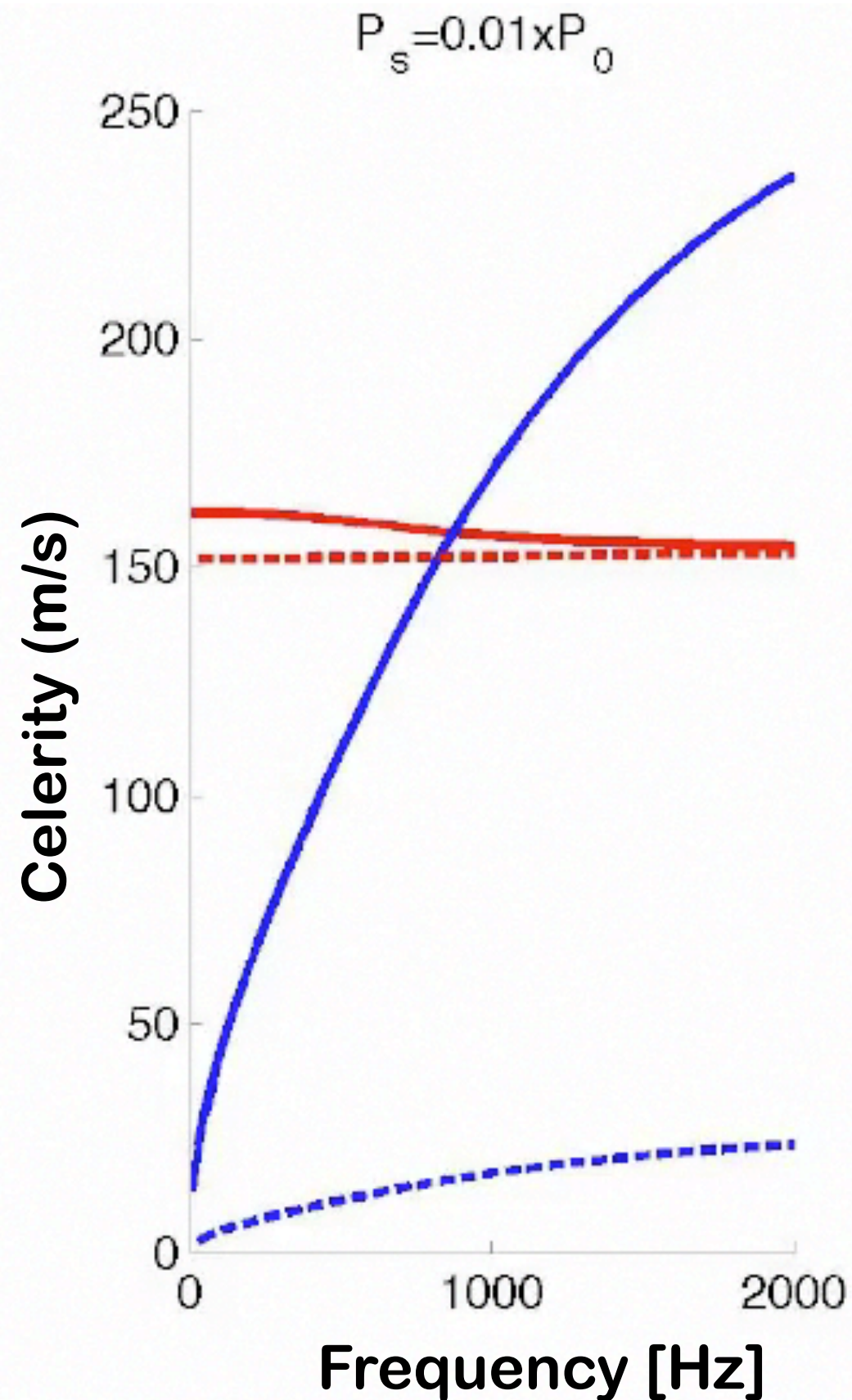
$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}}, \quad \delta_{s1} = \omega \sqrt{\frac{\tilde{\rho}}{\hat{P}}}, \quad \delta_{s2} = \omega \sqrt{\frac{\tilde{\rho}_s}{\hat{P}}}$$

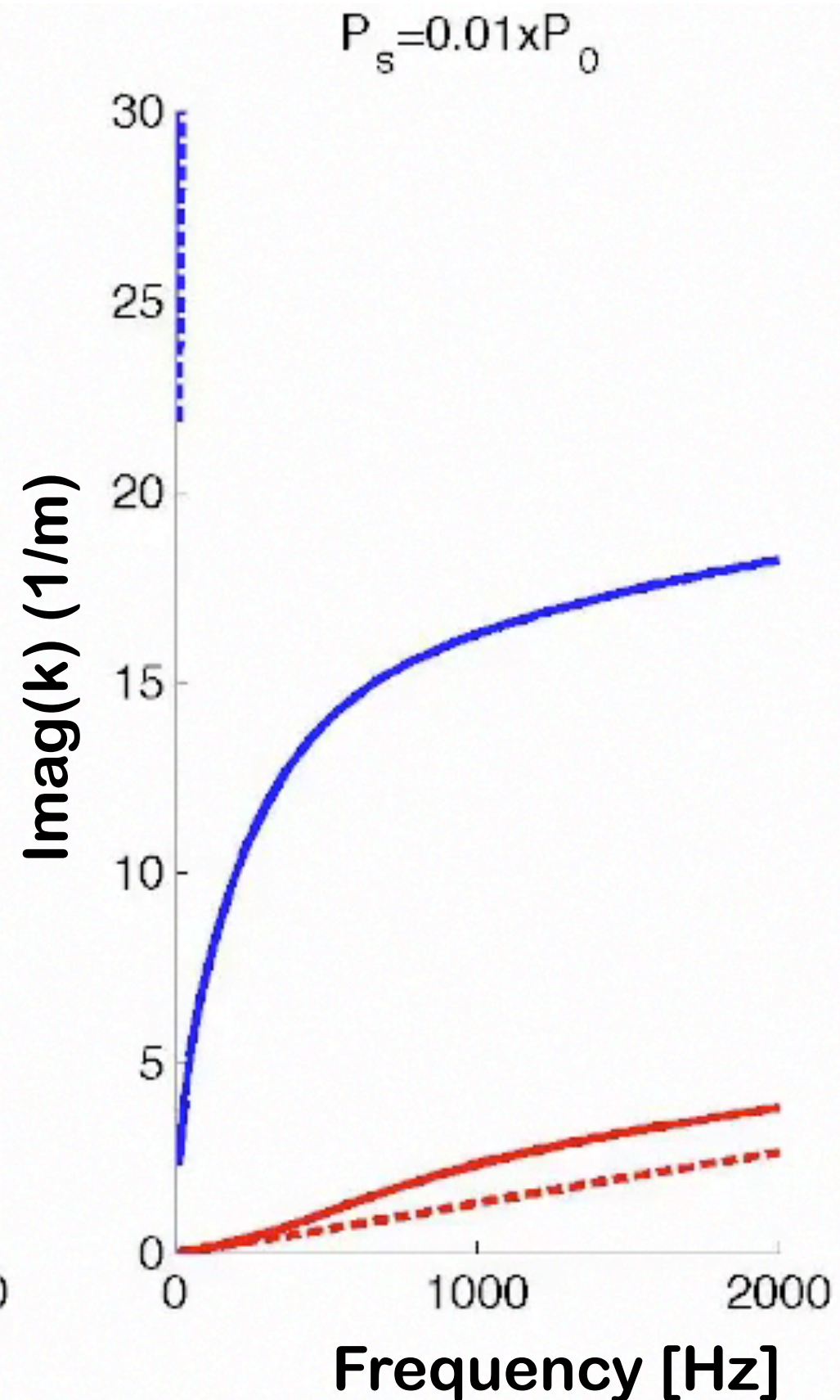
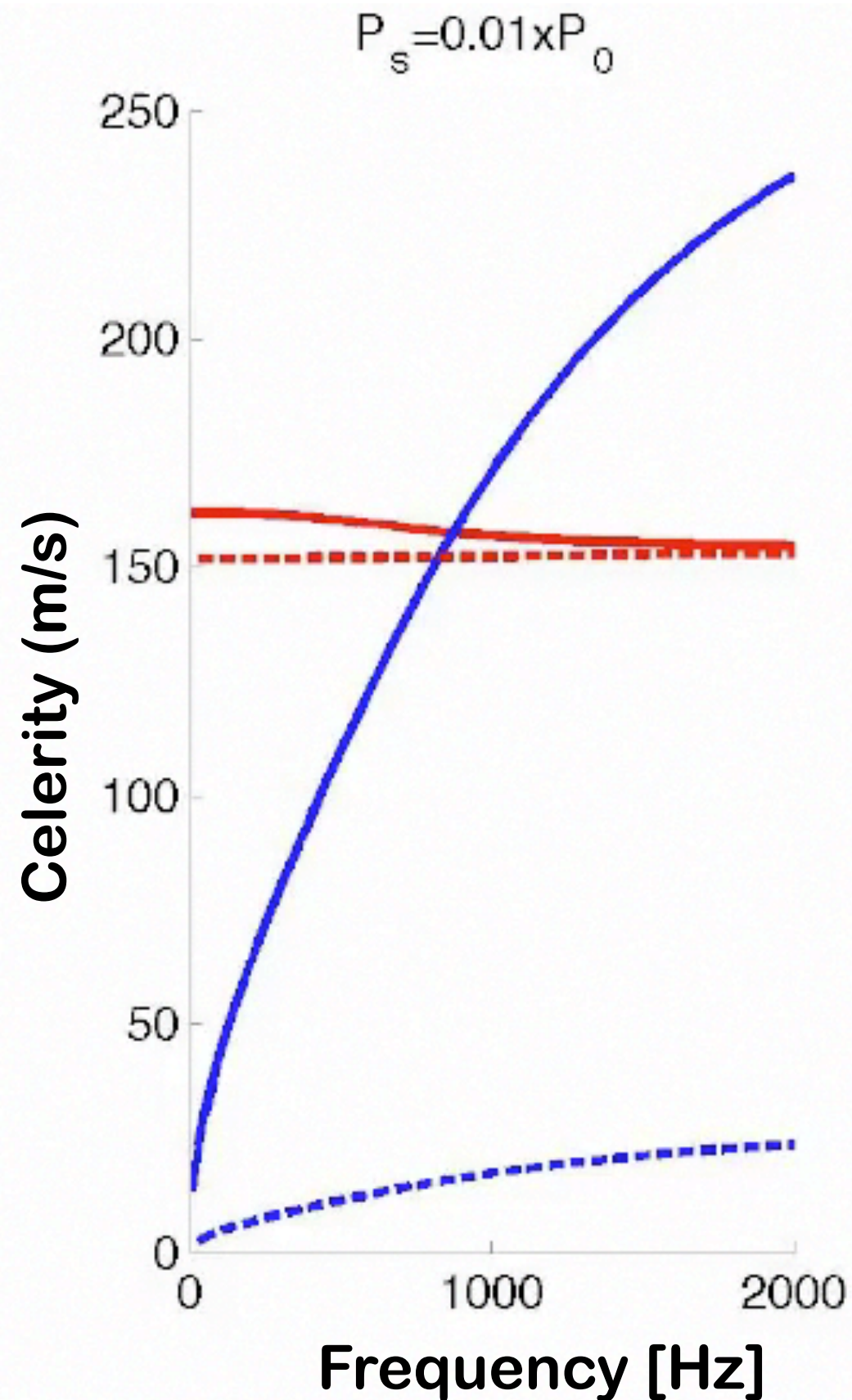
Amplitude ratio

$$\mu_i = \frac{\varphi^t}{\varphi^s} = \tilde{\gamma} \frac{(\delta_i^2 - \delta_{s2}^2)}{\delta_{s2}^2 - \delta_{s1}^2} = \tilde{\gamma} \frac{\delta_{eq}^2}{\delta_i^2 - \delta_{eq}^2}$$









$$\mathbf{u}^s = \nabla \wedge \boldsymbol{\psi}^s, \quad \mathbf{u}^t = \nabla \wedge \boldsymbol{\psi}^t$$

Gauge condition

$$\nabla \cdot \boldsymbol{\psi}^s = \nabla \cdot \boldsymbol{\psi}^t = 0$$

$$\mathbf{u}^s = \nabla \wedge \boldsymbol{\psi}^s, \quad \mathbf{u}^t = \nabla \wedge \boldsymbol{\psi}^t$$

Gauge condition

$$\nabla \cdot \boldsymbol{\psi}^s = \nabla \cdot \boldsymbol{\psi}^t = 0$$

$$-\omega^2 \begin{bmatrix} \tilde{\rho}_s & \tilde{\rho}_{eq} \tilde{\gamma} \\ \tilde{\rho}_{eq} \tilde{\gamma} & \tilde{\rho}_{eq} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\psi}^s \\ \boldsymbol{\psi}^t \end{Bmatrix} = \begin{bmatrix} N & 0 \\ 0 & 0 \end{bmatrix} \nabla^2 \begin{Bmatrix} \boldsymbol{\psi}^s \\ \boldsymbol{\psi}^t \end{Bmatrix}$$



6x6 system

$$\mathbf{u}^s = \nabla \wedge \boldsymbol{\psi}^s, \quad \mathbf{u}^t = \nabla \wedge \boldsymbol{\psi}^t$$

Gauge condition

$$\nabla \cdot \boldsymbol{\psi}^s = \nabla \cdot \boldsymbol{\psi}^t = 0$$

$$-\omega^2 \begin{bmatrix} \tilde{\rho}_s & \tilde{\rho}_{eq} \tilde{\gamma} \\ \tilde{\rho}_{eq} \tilde{\gamma} & \tilde{\rho}_{eq} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\psi}^s \\ \boldsymbol{\psi}^t \end{Bmatrix} = \begin{bmatrix} N & 0 \\ 0 & 0 \end{bmatrix} \nabla^2 \begin{Bmatrix} \boldsymbol{\psi}^s \\ \boldsymbol{\psi}^t \end{Bmatrix}$$



6x6 system

Singularity of the matrix

$$\mathbf{u}^s = \nabla \wedge \boldsymbol{\psi}^s, \quad \mathbf{u}^t = \nabla \wedge \boldsymbol{\psi}^t$$

Gauge condition

$$\nabla \cdot \boldsymbol{\psi}^s = \nabla \cdot \boldsymbol{\psi}^t = 0$$

$$-\omega^2 \begin{bmatrix} \tilde{\rho}_s & \tilde{\rho}_{eq} \tilde{\gamma} \\ \tilde{\rho}_{eq} \tilde{\gamma} & \tilde{\rho}_{eq} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\psi}^s \\ \boldsymbol{\psi}^t \end{Bmatrix} = \begin{bmatrix} N & 0 \\ 0 & 0 \end{bmatrix} \nabla^2 \begin{Bmatrix} \boldsymbol{\psi}^s \\ \boldsymbol{\psi}^t \end{Bmatrix}$$



6x6 system

Singularity of the matrix

Only one shear wave

$$\delta_s = \omega^2 \sqrt{\frac{\tilde{\rho}_s - \tilde{\gamma}^2 \tilde{\rho}_{eq}}{N}}$$

$$\mu_s = \frac{\boldsymbol{\psi}^t}{\boldsymbol{\psi}^s} = -\tilde{\gamma}$$

- **Three waves**
 - 1 compressional solid controlled
 - 1 compressional fluid controlled
 - 1 shear wave solid controlled
- Each wave is both solid and fluid

- Three waves
 - 1 compressional solid controlled
 - 1 compressional fluid controlled
 - 1 shear wave solid controlled
- Each wave is both solid and fluid

Expression of solid and total displacements

$$\nabla \wedge \mathbf{u}_1 = \nabla \wedge \mathbf{u}_2 = 0, \quad \nabla \cdot \mathbf{u}_s = 0$$

$$\mathbf{u}^s = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_s$$

$$\mathbf{u}^t = \mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \mu_s \mathbf{u}_s$$

- Three waves
 - 1 compressional solid controlled
 - 1 compressional fluid controlled
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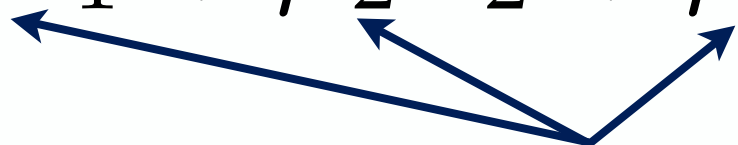
Expression of solid and total displacements

$$\nabla \wedge \mathbf{u}_1 = \nabla \wedge \mathbf{u}_2 = 0, \quad \nabla \cdot \mathbf{u}_s = 0$$

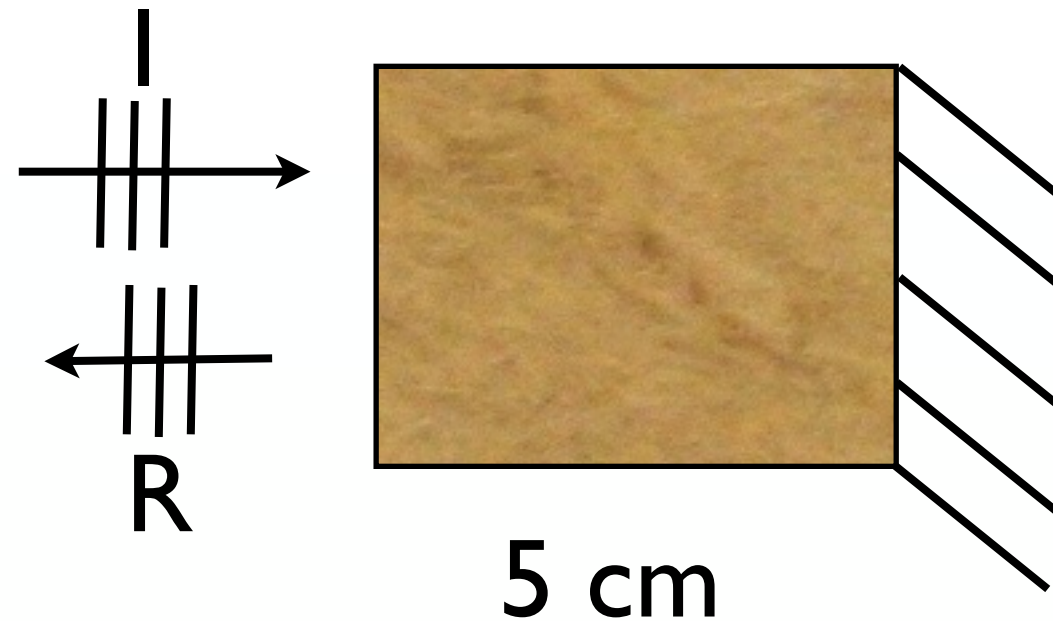
$$\mathbf{u}^s = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_s$$

$$\mathbf{u}^t = \mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \mu_s \mathbf{u}_s$$

Fixed amplitude ratio



Normal incidence impedance

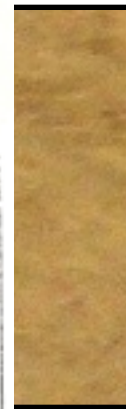


ϕ [1]	σ [$Nm^{-4}s$]	α_{∞} [1]	Λ [μm]	Λ' [μm]
0,97	87000	2.52	37	119
ρ_1 [kgm^{-3}]	E [kPa]	ν [1]	η_s [1]	
31	143	0.3	0.055	

PROPAGATION OF SOUND IN POROUS MEDIA

modelling sound absorbing materials

J.F. ALLARD



5

∞

2.5

ν

0.1

ρ_1

140

Propagation of Sound in Porous Media

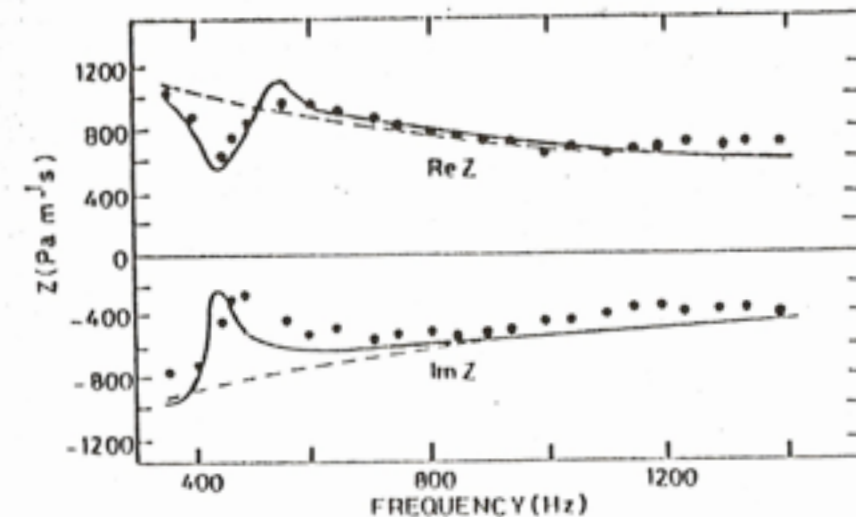
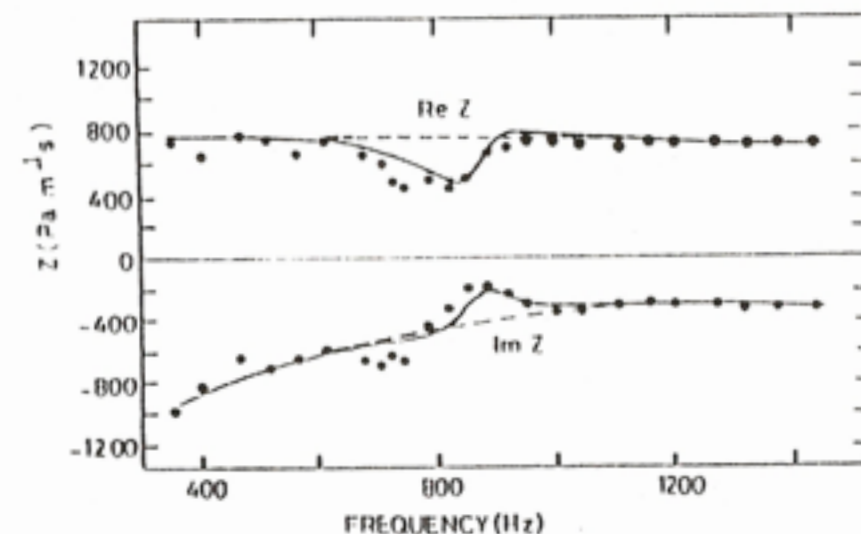


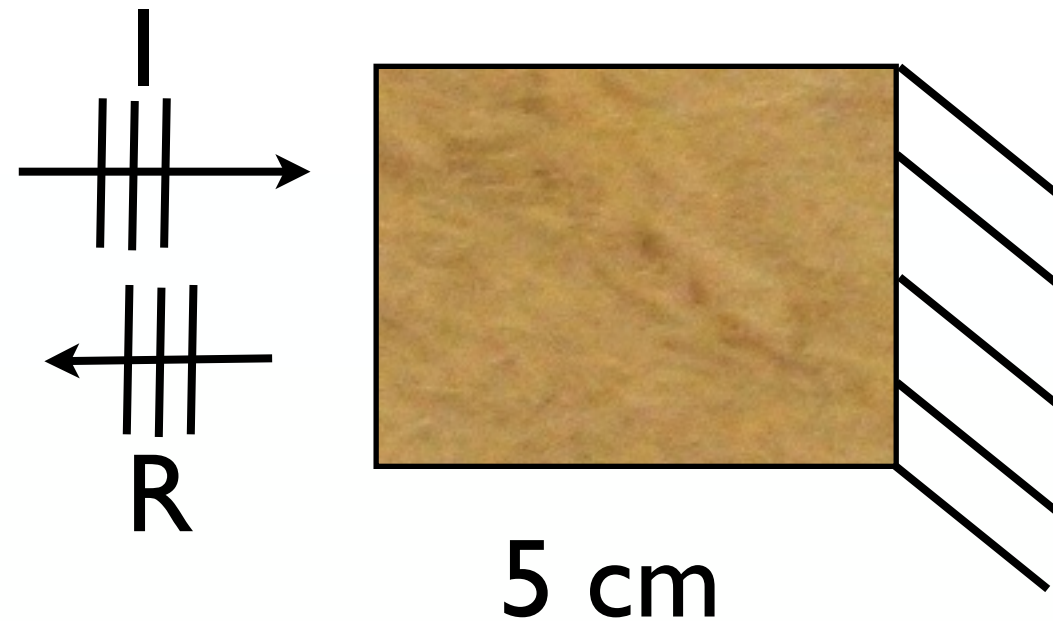
FIG. 6.10. Surface impedance Z of a layer of the fibrous material described in Section 6.5.4. The thickness of the layer is $l = 10$ cm. Prediction with eqn (6.108). —, Prediction for the same material with a rigid frame: ----. Measurements: ● ● ●. Measurements reproduced from Ref. [24].

surface impedances calculated by eqn (6.108), and by eqns (4.139) (6.87) and (6.88) for the same material with a rigid frame, are represented by the same curve except around the peaks which are not predicted by the one-wave model. The peaks appear around the $\lambda/$ resonance of the frame-borne wave which is located at the frequency

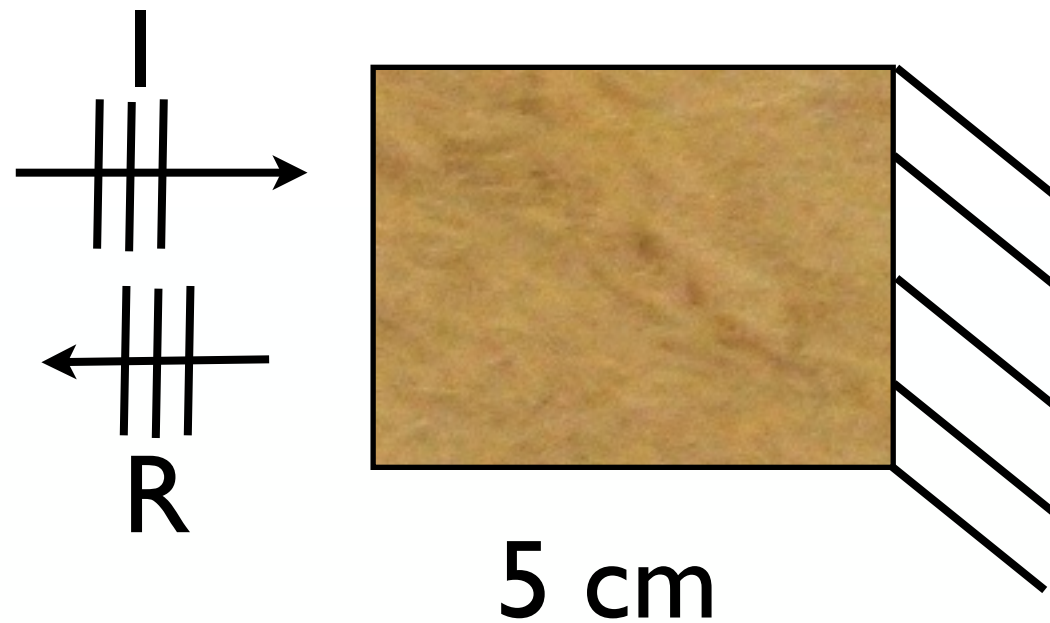


n

Normal incidence impedance



ϕ [1]	σ [$Nm^{-4}s$]	α_{∞} [1]	Λ [μm]	Λ' [μm]
0,97	87000	2.52	37	119
ρ_1 [kgm^{-3}]	E [kPa]	ν [1]	η_s [1]	
31	143	0.3	0.055	



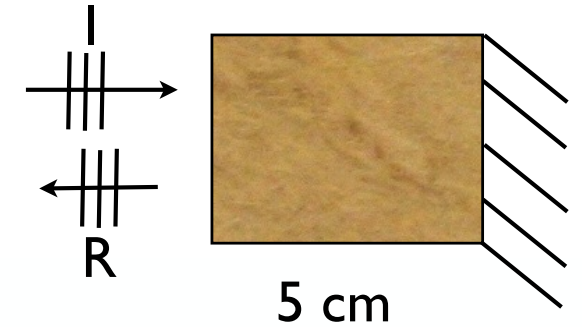
$$Z = \frac{\tilde{K}_{eq}}{j\omega} \times \frac{\delta_1 \delta_2 (\mu_2 - \mu_1)}{\delta_1 \mu_2 \tan(\delta_2 l) - \delta_2 \mu_1 \tan(\delta_1 l)}$$

$$u^s(x) = A \sin(\delta_1 x) + B \sin(\delta_2 x)$$

$$u^t(x) = \mu_1 A \sin(\delta_1 x) + \mu_2 B \sin(\delta_2 x)$$

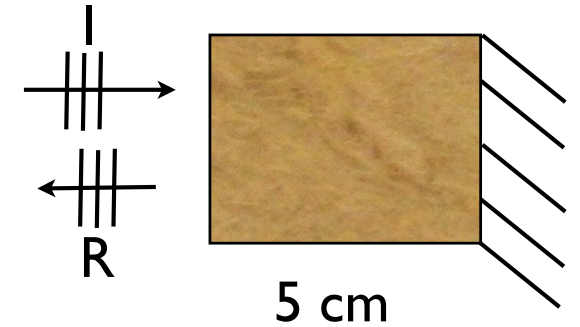
$$\hat{\sigma}^s(x) = \hat{P}[A \delta_1 \cos(\delta_1 x) + B \delta_2 \cos(\delta_2 x)]$$

$$p(x) = -\tilde{K}_{eq}[\mu_1 A \delta_1 \cos(\delta_1 x) + B \mu_2 \delta_2 \cos(\delta_2 x)]$$



$$Z = \frac{\tilde{K}_{eq}}{j\omega} \times \frac{\delta_1 \delta_2 (\mu_2 - \mu_1)}{\delta_1 \mu_2 \tan(\delta_2 l) - \delta_2 \mu_1 \tan(\delta_1 l)}$$

$$\begin{aligned}
 u^s(x) &= A \sin(\delta_1 x) + B \sin(\delta_2 x) \\
 u^t(x) &= \mu_1 A \sin(\delta_1 x) + \mu_2 B \sin(\delta_2 x) \\
 \hat{\sigma}^s(x) &= \hat{P}[A \delta_1 \cos(\delta_1 x) + B \delta_2 \cos(\delta_2 x)] \\
 p(x) &= -\tilde{K}_{eq}[\mu_1 A \delta_1 \cos(\delta_1 x) + B \mu_2 \delta_2 \cos(\delta_2 x)]
 \end{aligned}$$



at air-porous interface

$$0 = A \delta_1 \cos(\delta_1 l) + B \delta_2 \cos(\delta_2 l) \implies A = -\frac{\delta_2 \cos(\delta_2 l)}{\delta_1 \cos(\delta_1 l)} B$$

$$Z = \frac{\tilde{K}_{eq}}{j\omega} \times \frac{\delta_1 \delta_2 (\mu_2 - \mu_1)}{\delta_1 \mu_2 \tan(\delta_2 l) - \delta_2 \mu_1 \tan(\delta_1 l)}$$

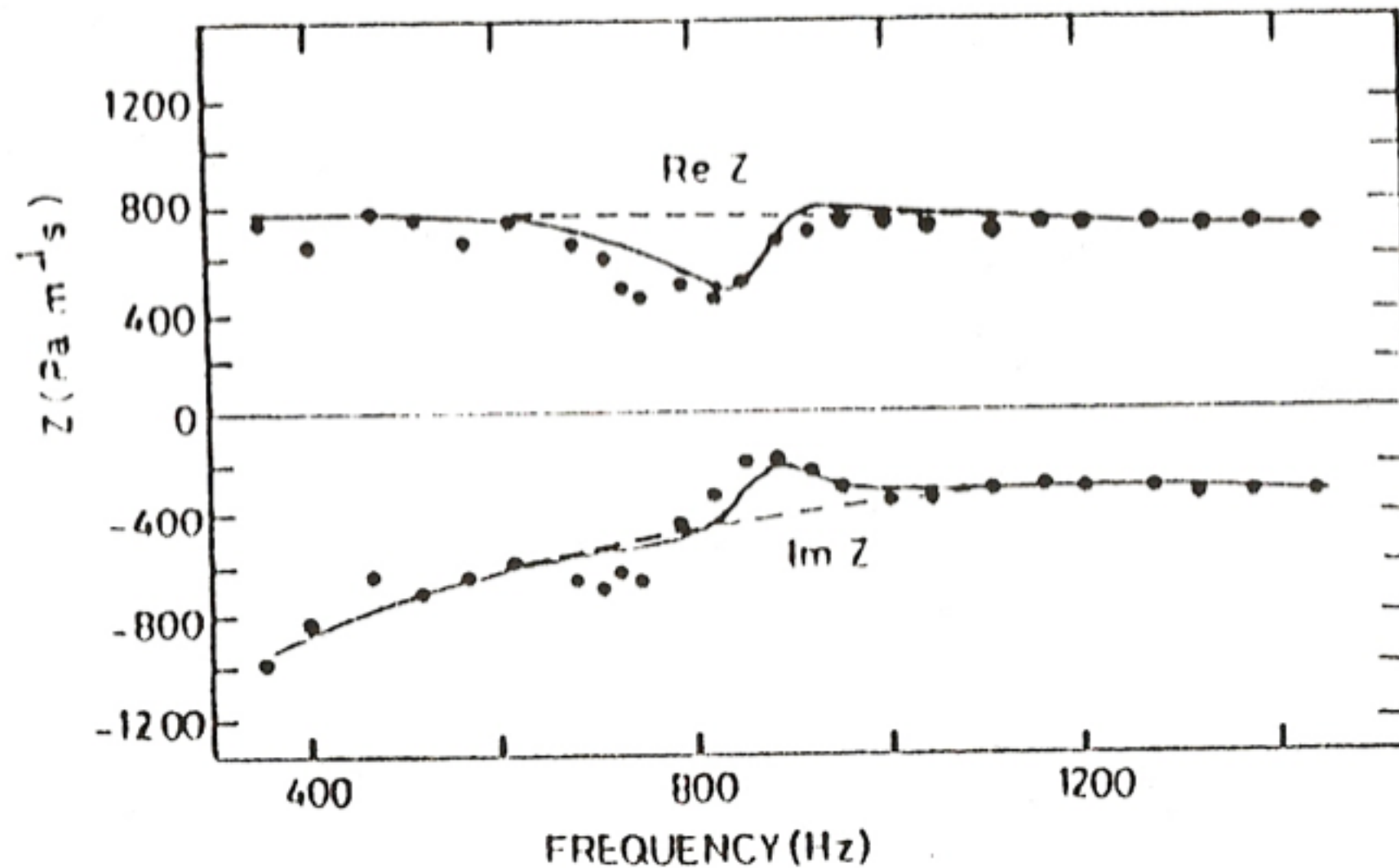
$$Z = \frac{\tilde{K}_{eq}}{j\omega} \times \frac{\delta_1 \delta_2 (\mu_2 - \mu_1)}{\delta_1 \mu_2 \tan(\delta_2 l) + \delta_2 \mu_1 \tan(\delta_1 l)}$$

$$Z = \frac{\tilde{K}_{eq}}{j\omega} \times \frac{\delta_1 \delta_2 (\mu_2 - \mu_1)}{\delta_1 \mu_2 \tan(\delta_2 l) + \delta_2 \mu_1 \tan(\delta_1 l)}$$

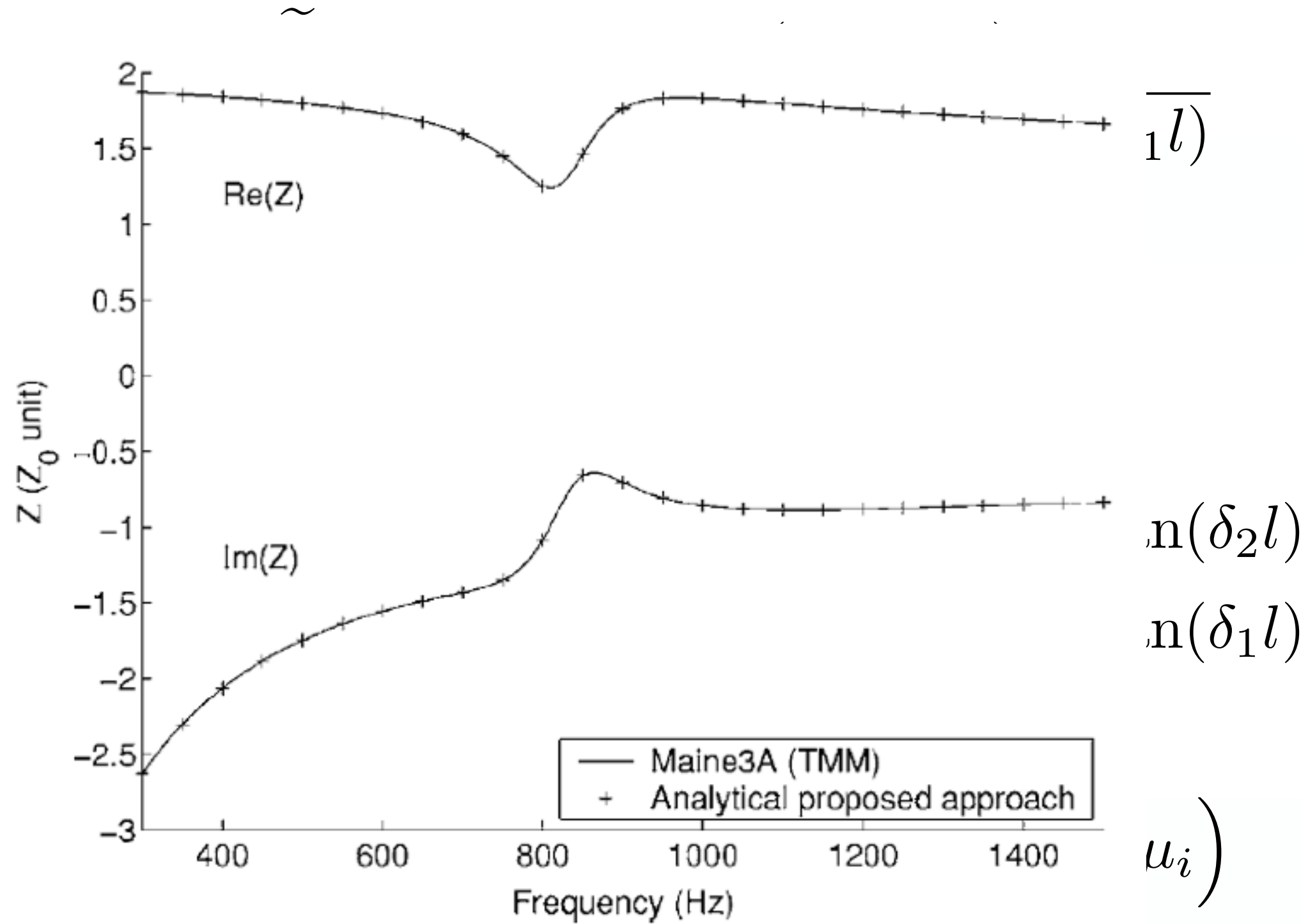
$$Z = -j \frac{Z_1^s Z_2^f \mu_2 - Z_2^s Z_1^f \mu_1}{D}$$

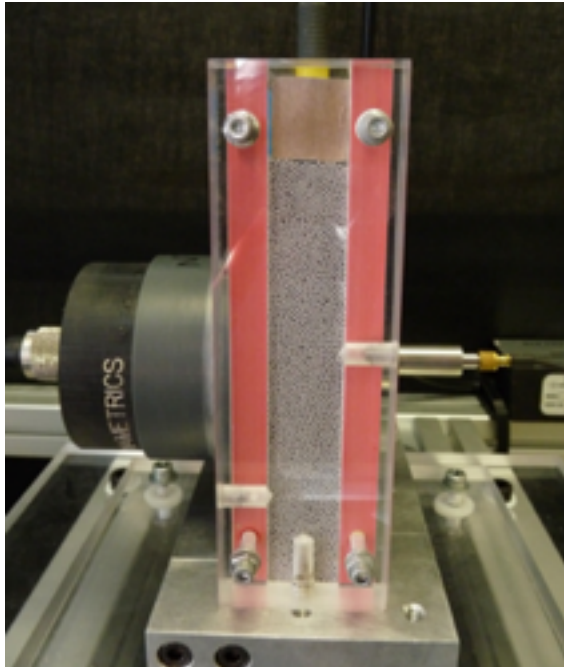
$$\begin{aligned} D &= (1 - \phi + \phi \mu_2) (Z_1^s - (1 - \phi) Z_1^f \mu_1) \tan(\delta_2 l) \\ &+ (1 - \phi + \phi \mu_1) (Z_2^s - (1 - \phi) Z_2^f \mu_2) \tan(\delta_1 l) \end{aligned}$$

$$Z_i^f = \frac{\delta_i}{\phi \omega} \left(\tilde{R} + \frac{\tilde{Q}}{\mu_i} \right), \quad Z_i^s = \frac{\delta_i}{\omega} \left(\tilde{P} + \tilde{Q} \mu_i \right)$$



$$Z_i^f = \frac{\delta_i}{\phi\omega} \left(\tilde{R} + \frac{Q}{\mu_i} \right), \quad Z_i^s = \frac{\delta_1}{\omega} \left(\tilde{P} + \tilde{Q}\mu_i \right)$$





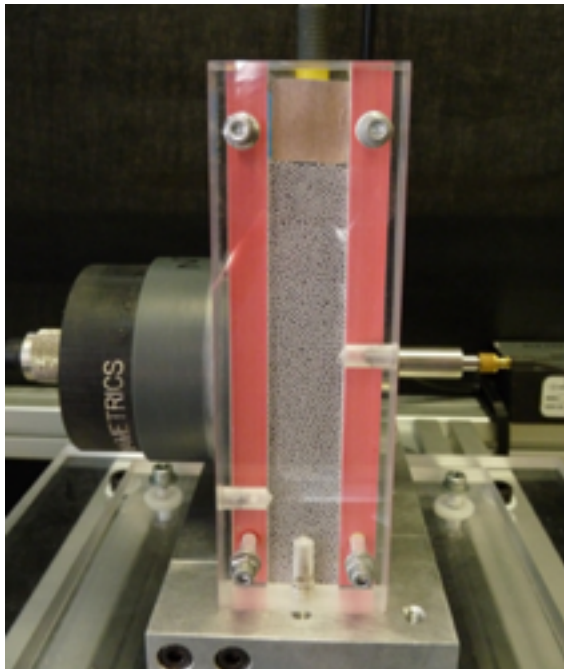
Transmission of a 8 cm glass bead layer ($d=2$ mm)

Transmission(dB)

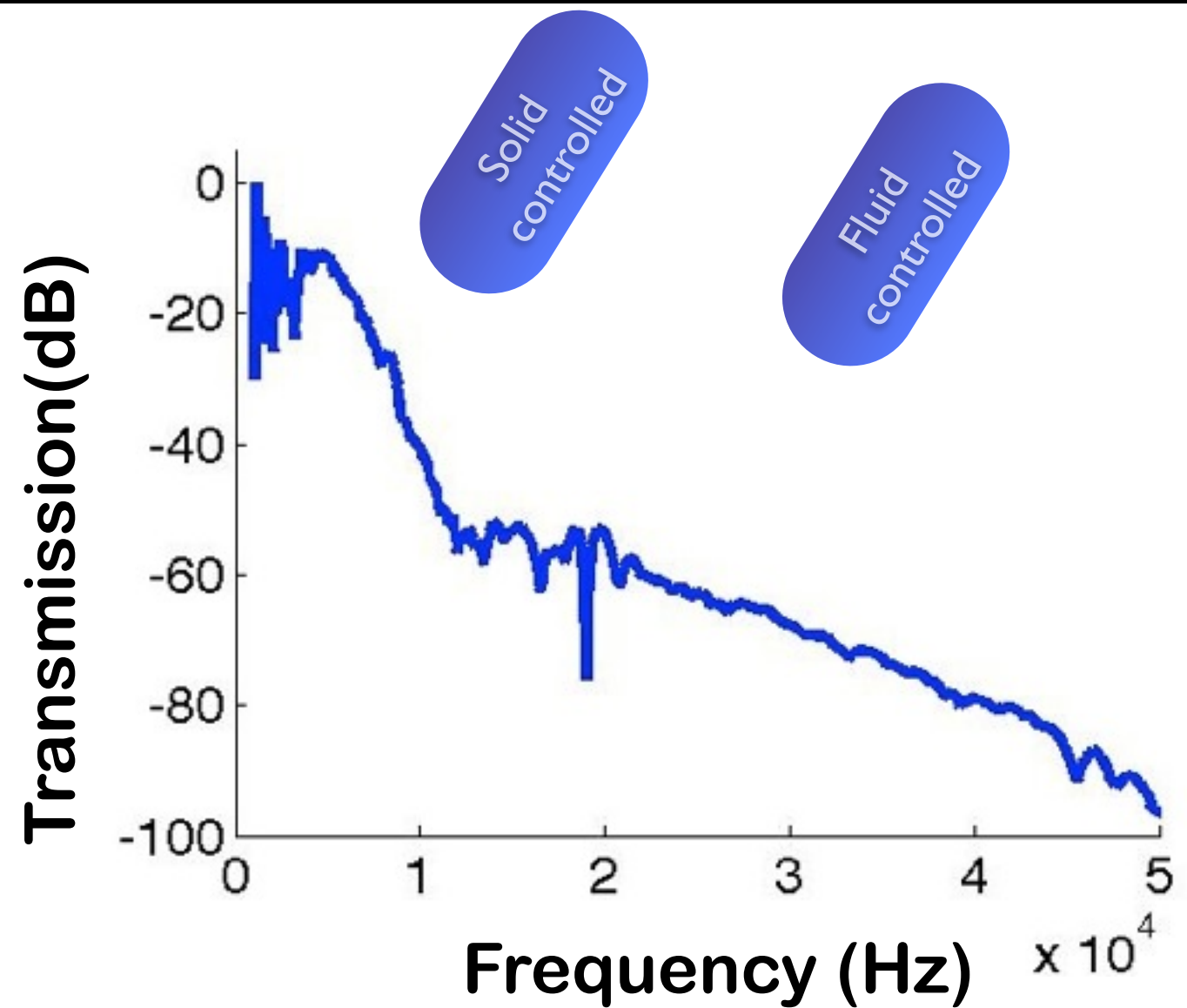
Solid controlled

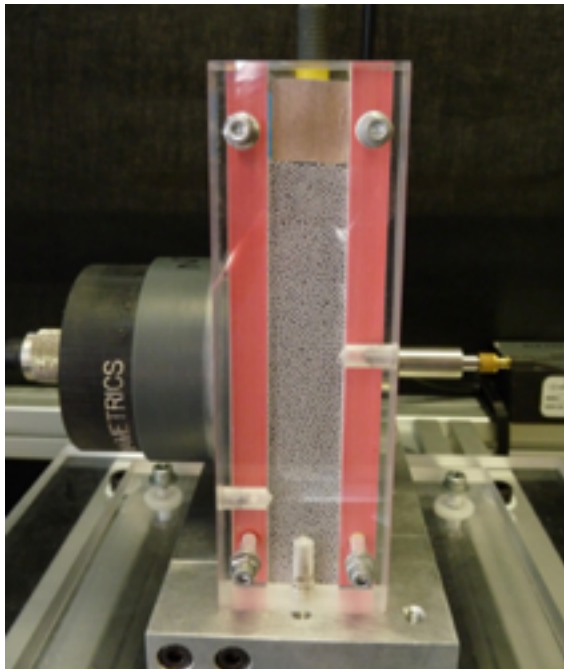
Fluid controlled

Frequency (Hz)

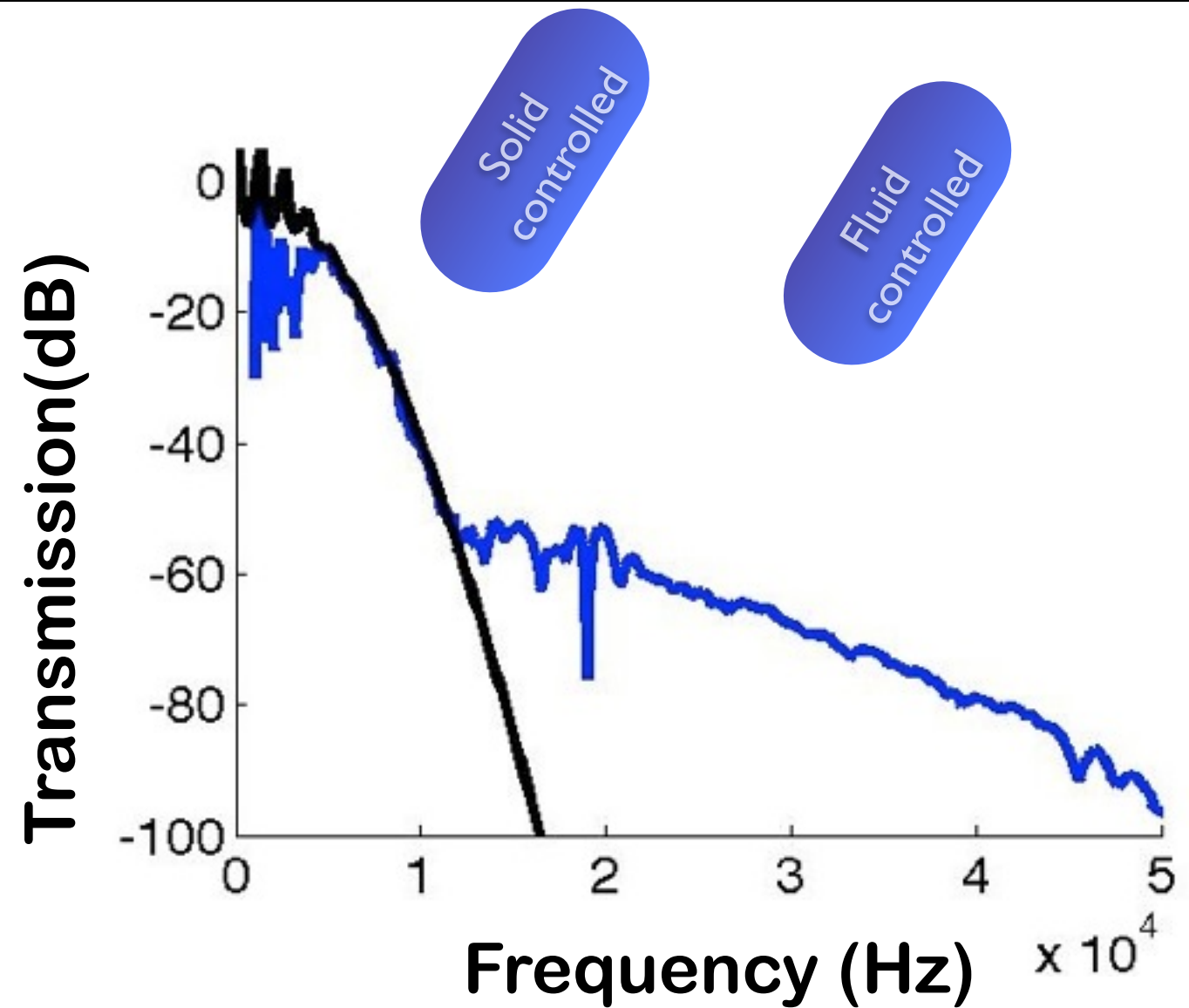


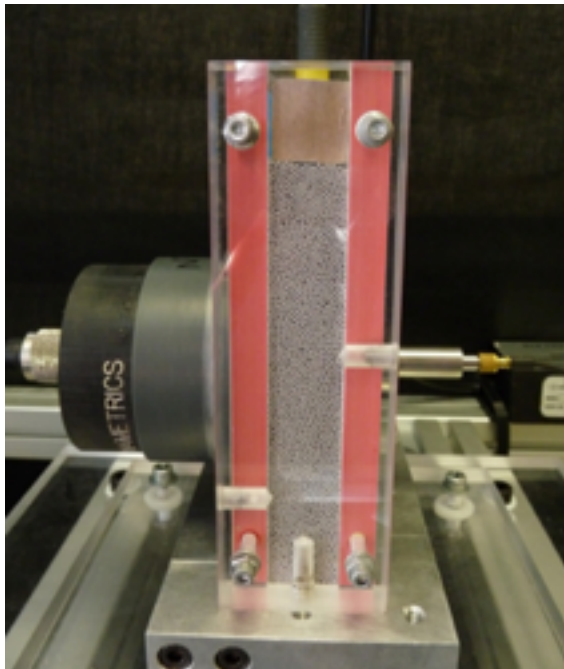
Transmission of a 8 cm glass bead layer ($d=2$ mm)



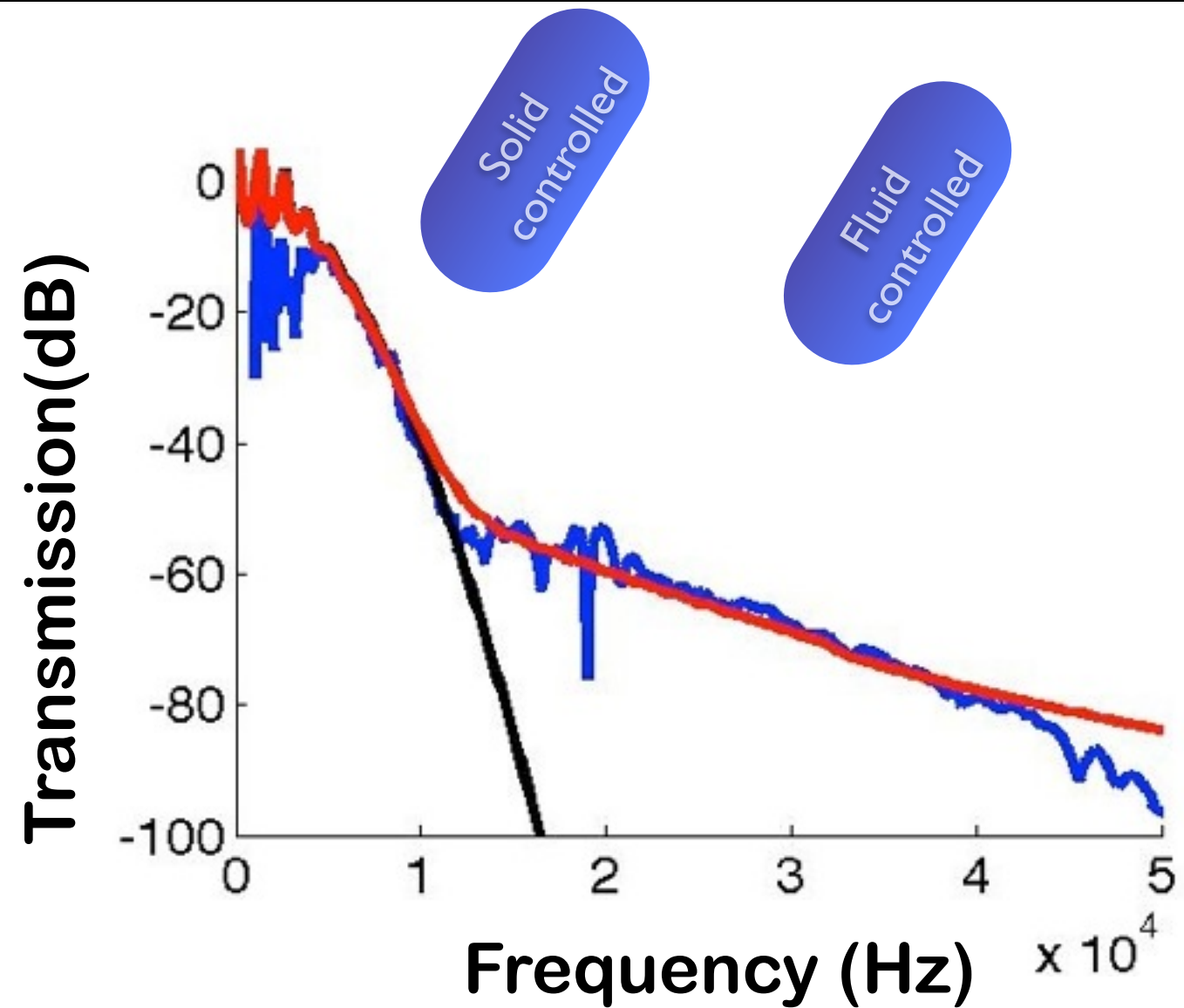


Transmission of a 8 cm glass bead layer ($d=2$ mm)

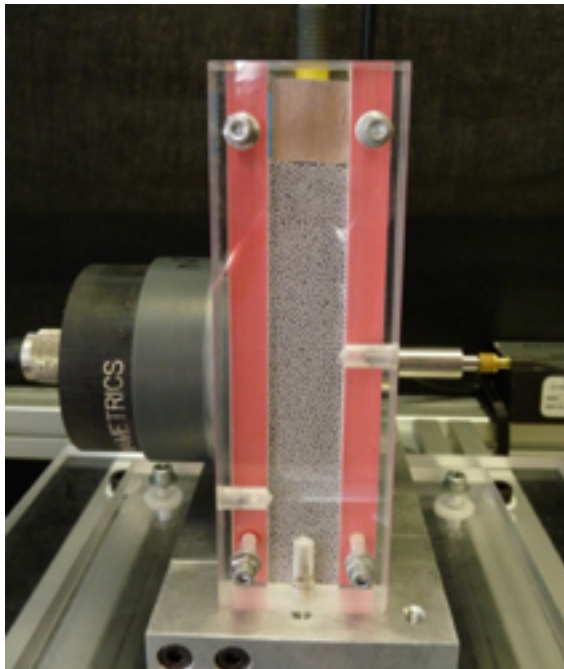




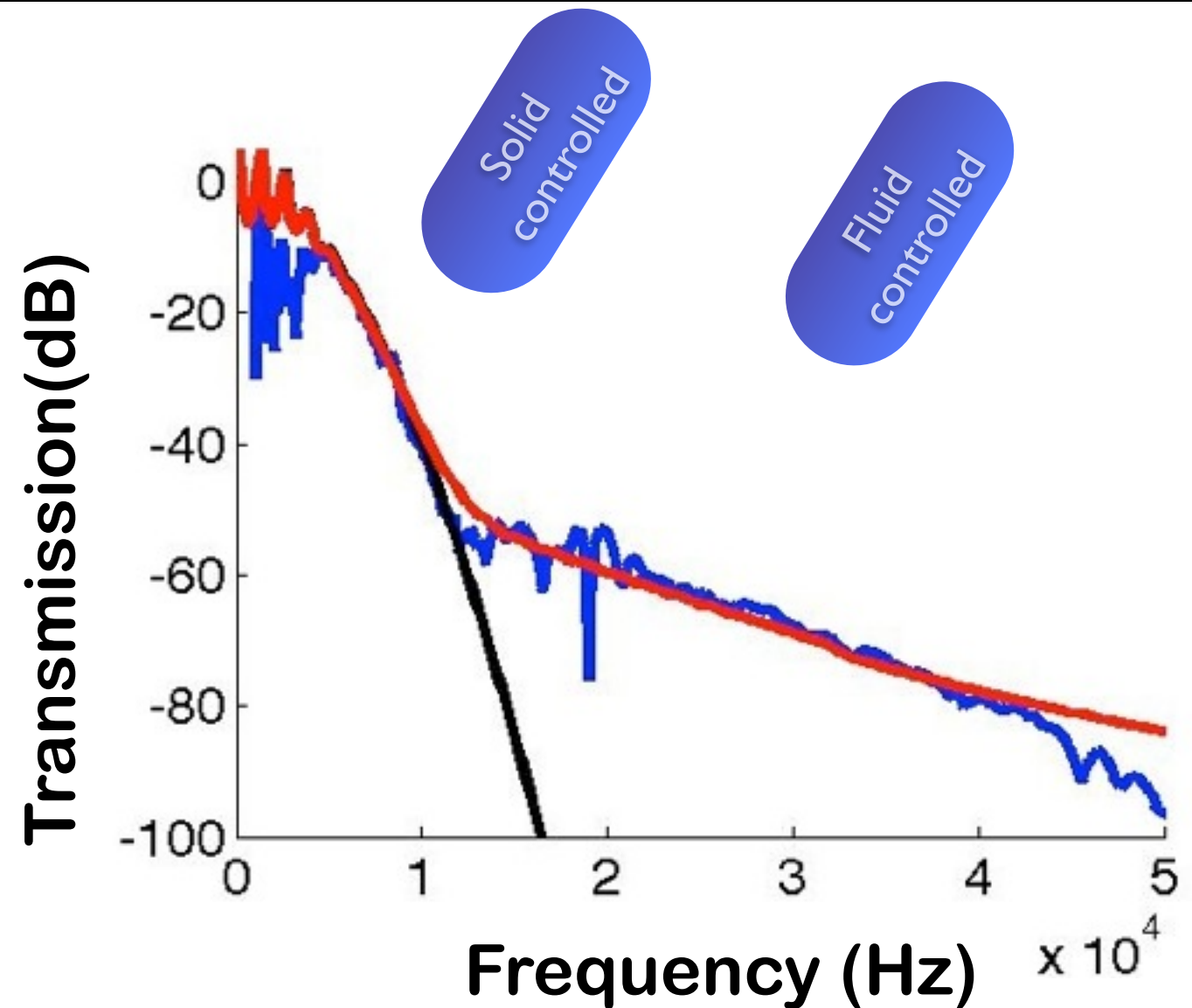
Transmission of a 8 cm glass bead layer ($d=2$ mm)



In-vacuo model $\nabla \cdot \sigma_{nl}(u^s) = -\omega^2 \rho_s u^s$



Transmission of a 8 cm glass bead layer ($d=2$ mm)

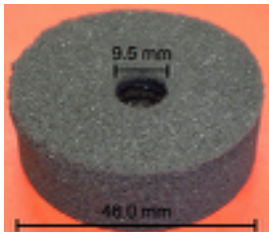


In-vacuo model $\nabla \cdot \sigma_{nl}(u^s) = -\omega^2 \rho_s u^s$

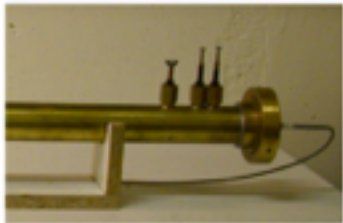
$$\nabla \cdot \hat{\sigma}_{nl}(u^s) = -\omega^2 \tilde{\rho}_s u^s - \omega^2 \tilde{\rho}_{eq} \tilde{\gamma} u^t,$$

$$\tilde{K}_{eq} \nabla \nabla \cdot u^t = -\omega^2 \tilde{\rho}_{eq} \tilde{\gamma} u^s - \omega^2 \tilde{\rho}_{eq} u^t.$$

Equivalent fluid model $\tilde{K}_{dp} \nabla^2 p + \omega^2 \tilde{\rho}_{dp} p = 0$



Matelys AcV



Absorption coefficient

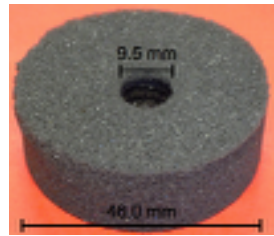
Frequency (Hz)

Exp	Eq. fluid	Biot DP

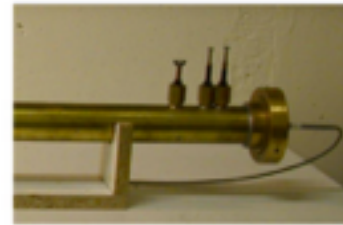
Equivalent fluid model $\tilde{K}_{dp} \nabla^2 p + \omega^2 \tilde{\rho}_{dp} p = 0$

$$P_{dp} \frac{\partial^2 u^s}{\partial x^2} = -\omega^2 \tilde{\rho}_s^{dp} u^s - \omega^2 \tilde{\gamma}^{dp} \tilde{\rho}_{dp} u^t,$$

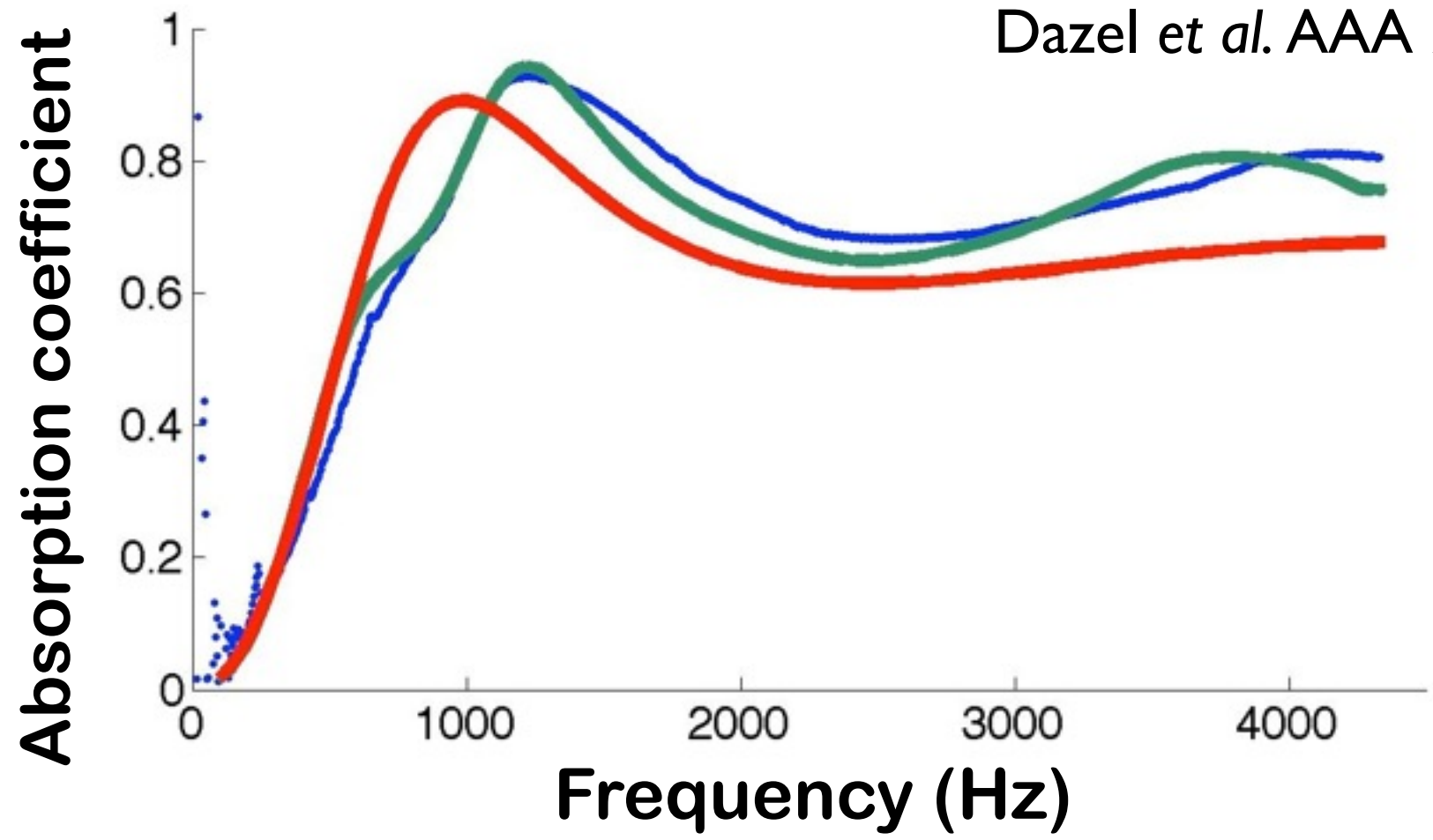
$$\tilde{K}_{dp} \frac{\partial^2 u^t}{\partial x^2} = -\omega^2 \tilde{\gamma}^{dp} \tilde{\rho}_{dp} u^s - \omega^2 \tilde{\rho}_{dp} u^t.$$



Matelys AcV



Dazel et al. AAA 2012



Exp	Eq. fluid	Biot DP